

The Dismal State of the Dismal Science in South Africa*

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Abstract

Researchers based in South Africa rarely publish in the top economics journals. Across 1990–2025, only 2 regular top-five articles and 11 in a specified ten-journal set list a South African institution as an author’s first affiliation. Chile, Brazil, Argentina and Colombia have substantially more; holding author numbers equal preserves the gap with Chile. Counting economists educated in South Africa, wherever they work, adds few articles: the identified group has six regular top-five articles, against 108 for Chile. South Africa’s top-five share ranks 24th among 26 countries under four journal denominators, although recent-period ranks vary with the denominator. Between 2000–09 and 2015–24, recorded domestic output grew 4.5 times, but growth within sources represented in the base period was 1.27 times. Among leading-journal authors, prior collaborations with top-five authors are less common in South Africa than in four Latin American peers, even after matching journal and period. Finance is the largest topic in domestic research, while leading-journal research about South Africa concentrates on households. The findings and external evidence on training and networks motivate support for leading doctoral placements and sustained collaboration within domestic research careers.

Keywords: economics profession; South Africa; bibliometrics; research incentives; talent allocation; administrative data

JEL codes: A11; A14; I23; O55

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1 Introduction

Researchers based in South Africa rarely publish in the top economics journals. Across 1990–2025, only 2 regular articles in the conventional top five have an author whose first affiliation is South African. Both appeared in 2022. Chile has 30, Brazil 39, Czechia 21, Argentina 21 and Colombia eight under the same journal and affiliation rules. The difference is not simply one of author numbers: drawing a South Africa-sized pool from Chile still yields many more top-five contributions and repeat authors in the wider journal set.

This small domestic presence occurs in a country with several internationally recognised universities. South Africa has 4 universities in the Times Higher Education (THE) global top 500 for 2026, compared with two in Brazil and one each in Chile and Czechia (Times Higher Education, 2025). Overall rankings describe the wider institutions in which economists work; they do not measure economics staffing, departmental quality or expected publication. Reputation surveys carry a third of the THE weight (Times Higher Education, 2026), and ranking indicators largely reduce to reputation and total research output (Selten et al., 2020), so a well-known national university can rank above institutions with stronger economics departments. The closer comparison is between researchers publishing under common journal and affiliation rules. Why do so few economists at South African institutions contribute to the discipline’s leading journals?

In 1966 the *South African Journal of Economics* asked whether the country was training enough economists (Truu, 1966). The profession has since grown, but more economists and more publications need not mean a larger contribution to international research. This paper examines who produces economics research in South Africa, who studies the country and which questions they pursue. It measures domestic production through authors’ affiliations at publication, international attention through leading-journal articles about South Africa, and topic composition against comparable countries and government speeches. The publication measure captures research produced, rather than every person’s capacity to produce it. Top-five publication strongly influences academic careers in economics, although it is a noisy measure of the quality of individual papers (Heckman and Moktan, 2020); the paper therefore reports wider journal sets and citation impact alongside it. Nor is it the only valuable output: much South African research informs domestic policy without appearing in these journals.

The weakness at the top extends beyond the conventional five journals. Adding five general economics journals raises South Africa’s domestic count to 11, compared with 98 in Chile, 85 in Brazil, 42 in Argentina and 27 in Colombia. The full-period top-five share ranks 24th of 26 under the nineteen-journal denominator and after excluding the two development outlets that dominate South Africa’s output. In 2020–24 the absolute counts remain below Chile, Brazil, Czechia and Colombia, although the share rank is more sensitive to the journal basket.

Domestic journal output grew 4.5 times between 2000–09 and 2015–24, while publication in the nineteen journals grew 2.4 times, from a base of just 26 article-equivalents. Part of the output growth reflects changes in the journals represented in the database: restricting each country’s output to source IDs observed in its 2000–09 records reduces South Africa’s output growth to

1.27 times and changes its relative placement-growth rank from 6 to 10 of 26. Citation impact kept pace with recorded output. Across 1990–2025, 221 authors published with a South African first affiliation in the journal set. A comparison holding author numbers equal finds fewer repeat contributors and fewer top-five positions than in Chile.

Where South African economists work explains little of the small count. A country’s contribution to economics also includes the economists it educates who build careers abroad (Xie and Freeman, 2020). Counting articles by country of education rather than affiliation, the 111 identified South African-educated authors have six regular top-five articles in 1990–2025, five of them with such an author working abroad. Chilean-educated authors have 108, Colombian-educated 42 and Argentine-educated 233. The identification frame omits some economists abroad in every country, but such omissions are unlikely to account for an eighteenfold difference with Chile.

Research about South Africa has attracted more foreign than domestic authorship. In the 244 articles about the country in nineteen leading journals, foreign-based authors hold at least 57 per cent of fractional authorship in every period. The domestic share rose from 14 per cent in 1995–99 to 33 per cent in 2015–19, largely through *World Development* and the *Journal of African Economies*. These two journals carry 64.8 per cent of articles about South Africa and account for 86.3 per cent of domestic fractional output across all subjects in the nineteen-journal set. The broader journal total thus conceals how little domestic authors publish at the highest tier.

Research produced in South Africa and leading-journal work about the country also address different questions. Finance is the largest domestic topic, at 20.5 per cent of output. Household research accounts for 4.02 times its predicted share in leading-journal work about South Africa. Alternative income, population and period benchmarks preserve the concentration on finance and households. Firm-topic shares are below prediction in both portfolios, but classification uncertainty makes the size of this difference less secure. Household surveys helped researchers study the country; linked tax records later supported work on firms, workers and trade. The domestic share of firm research nevertheless changed little relative to peers across the mid-2010s. This comparison tracks published topics; it does not count uses of the tax data or estimate the effect of opening them.

The publication records also show fewer connections to established top-five authors. Among domestic author positions in the nineteen journals during 2015–24, 7.0 per cent have a recorded prior collaboration with a top-five author, against 14.3–34.4 per cent in Colombia, Brazil, Chile and Argentina. Both the collaboration and the partner’s top-five publication precede the article being counted. Matching the four peers to South Africa’s journal-period mix leaves a gap. Among eventual leading-journal authors with known doctoral histories, the South African-educated group also has a smaller foreign-doctoral share than Uruguay, Chile, Argentina and Colombia. These selected-author comparisons describe training routes and recorded relationships. Together with external evidence on faculty quality and networks, they motivate preparing more promising students for leading doctoral departments and supporting their research careers afterwards.

The paper builds on research that distinguishes work about developing countries from work produced there (Das et al., 2013; Porteous, 2022; Chelwa, 2021; Amarante et al., 2022; Amarante

and Zurbriigg, 2022; Aigner, Greenspon and Rodrik, 2025). Seekings (2001) noted early in the democratic period that much new quantitative research on South Africa was produced abroad. This paper measures both for South Africa over time and compares domestic publication across countries under common journal and affiliation rules. Studies of South African publication incentives (Luiz, 2009; Muller, 2017; Mouton and Valentine, 2017; Tomaselli, 2018; Kerr and de Jager, 2021; Yu, Kasongo and Moses, 2017) and count-based funding elsewhere (Butler, 2003; Hicks, 2012; Tonta and Akbulut, 2020) explain why publication volume and journal placement should be examined separately. Here, volume outgrows placement in most peers in the unrestricted records, but that result changes when the source set is held fixed. Neither comparison identifies the effect of South Africa’s subsidy. Evidence on faculty quality and academic networks (Waldinger, 2010; Colussi, 2018; Carrell, Figlio and Lusher, 2024) informs the training and collaboration comparisons. Work on administrative data (Card et al., 2010; Einav and Levin, 2014) informs the analysis of topics, which relates changes in research to the development of household and firm data while documenting published applications (Pieterse, Gavin and Kreuser, 2018; Fedderke, Obikili and Viegi, 2018).

Sections 2–4 describe the historical setting, framework and data. Sections 5–9 report the publication, topic and researcher comparisons. Sections 10 and 11 examine possible explanations and research policy.

2 Historical background

The Economic Society of South Africa was founded in 1925, and the *South African Journal of Economics* followed in 1933 (Pearsall, 1939). Truu’s 1966 note was about labour supply: did the universities produce enough economists for government and business? (Truu, 1966). Universities and domestic journals expanded thereafter, but apartheid and the academic boycott limited contact with the international profession. The changing subjects and methods of the Society’s journal are documented by Snowball and Kramm (2026). The analysis here follows South African participation in international research after 1990.

The base from which the post-1994 profession grew was small and was built to exclude most of the population. University segregation was enforced by law from 1959 (Union of South Africa, 1959), and the academic economics profession of 1990 was almost entirely white; the universities reserved for black South Africans had little research economics; and the academic boycott of the 1980s restricted conference travel, visiting appointments and co-authorship more than it restricted access to published work. Apartheid thus restricted both entry into economics and the international relationships through which researchers developed their work. Skilled emigration after 1994 is documented (Stern and Szalontai, 2006), but the South African-educated diaspora holds only 3 to 14 per cent of leading-journal authorship about the country in any period (Section 5); this share describes the authorship of work about South Africa, rather than the fraction of the diaspora’s own output devoted to the country.

The apartheid state quantified a great deal for administrative purposes. Population censuses,

wage statistics by race and sector and manpower surveys were produced by a bureaucracy that employed demographers and statisticians, and an earlier inquiry into the “poor white problem” had combined economic, psychological, educational and sociological investigation (Carnegie Commission, 1932). Yet the state collected little usable information on the incomes and unemployment of African South Africans, so economists who studied poverty or unemployment in the whole population had to infer them as residuals from aggregate data (Seekings, 2001). Most of this quantitative work came from liberal political economists. The Marxist and revisionist scholarship that dominated much of South African social science from the 1970s, including the social history associated with the Wits History Workshop, favoured theory-driven or qualitative research and paid little attention to the quantitative work of its liberal opponents (Seekings, 2001). The Second Carnegie Inquiry into Poverty and Development of the 1980s assembled many local studies rather than national estimates (Wilson and Ramphele, 1989). Quantitative economics was thus the method of one camp in these debates.

The political transition restored opportunities for international collaboration and raised scholarly interest in South Africa. The 1993 Project for Statistics on Living Standards and Development was the first countrywide household income and expenditure survey to cover the whole population. It was conducted shortly before the first democratic election, largely at the request of the African National Congress, and the National Income Dynamics Study added a household panel in 2008. Policy demand for quantitative analysis grew quickly, but much of the new research on the country was conducted by scholars based abroad (Seekings, 2001).

The national research funding formula pays universities a fixed unit for articles in accredited journals. A per-article subsidy has existed since the 1980s; the 2003 policy revised its weights and coverage (Department of Higher Education and Training, 2015; Muller, 2017), so the 1990s base formed under a count-based reward as well. Critics describe the subsidy as rewarding volume over quality (Tomaselli, 2018). Publication counts rose (Luiz, 2009; Yu, Kasongo and Moses, 2017), and part of the increase occurred in predatory outlets (Mouton and Valentine, 2017; Kerr and de Jager, 2021).

Firm-level tax data became available to researchers much later than household surveys. A National Treasury–UNU-WIDER call invited proposals in 2014; panel documentation and empirical working papers followed in 2016, and a special issue of the *South African Journal of Economics* appeared in 2018 (UNU-WIDER, 2014; Pieterse, Gavin and Kreuser, 2018; Fedderke, Obikili and Vieg, 2018). Access developed through research projects and a secure facility. Researchers gained access project by project, rather than through a single opening of linked company-tax, payroll and customs records to the entire profession.

3 Conceptual framework

Consider researcher i , with research ability a_i and international network n_i , choosing a topic k and a publication tier q . Expected payoff is

$$\pi_i(k, q) = p_k(q \mid a_i, n_i, D_k) (b + \rho_q) + d_k + \varepsilon_{ik} - c(q, a_i, n_i) - f_k(D_k). \quad (1)$$

The publication probability p_k depends on ability, networks and the data infrastructure D_k . The term b is the return common to accredited articles; ρ_q is the additional return to placement at tier q . Topic-specific demand, including grants and policy interest, is d_k ; ε_{ik} captures idiosyncratic preferences. The effort cost of a given tier is c , and $f_k(D_k)$ is the topic's fixed data cost. Better data reduce f_k . Let w_i^A denote baseline academic compensation, separate from the research-specific returns in π_i . The researcher remains in domestic academia when academic compensation and the best research payoff together exceed the outside wage:

$$w_i^A + \max_{k,q} \pi_i(k, q) \geq w_0 + \beta a_i, \quad (2)$$

where $w_0 + \beta a_i$ represents finance, consulting, government employment or emigration.

Lower data costs make a topic more attractive, holding demand, researcher characteristics and other opportunities fixed. The household-survey timing in Section 5 is consistent with that mechanism. The firm-data comparison in Section 8 asks whether the domestic portfolio shifted as another source of data became available. Data availability lowers a researcher's costs only if that researcher can obtain and use the records.

A larger common reward b favours the more readily attainable publication tier when its probability of acceptance is higher. Whether that produces unusually rapid growth in publication count depends on the wider research system. Section 7 therefore compares South Africa's volume-placement gap with those of other countries, and uses citation impact as a separate measure. A growth comparison can distinguish an unusual trajectory; it cannot identify the effect of a subsidy that already existed in the base period.

The participation condition describes who stays in research. If the outside wage net of baseline academic compensation rises with ability faster than the best research payoff, the most able may leave. Research productivity is also highly concentrated among economics PhD graduates, including those from leading departments; programme rank alone predicts individual output poorly (Conley and Önder, 2014). The size-matched comparison in Section 7 asks whether South Africa has fewer prolific authors than an equally sized group elsewhere. The talent-allocation and finance-wage literature (Murphy, Shleifer and Vishny, 1991; Philippon and Reshef, 2012; Célérier and Vallée, 2019) suggests how pay could affect career choices, but publication counts alone cannot establish who left academia.

Networks can raise publication probabilities and reduce research costs. A leading doctoral department supplies advanced training, repeated feedback and contact with researchers who shape

the international research agenda. Faculty quality affects doctoral outcomes (Waldinger, 2010), and editor–author connections are associated with publication opportunities in economics (Colussi, 2018). Training and networks are potentially complementary: better training makes demanding collaborations feasible, while those collaborations raise the return to training. The loss of productive collaborators lowers research output in the life sciences (Azoulay, Graff Zivin and Wang, 2010), providing evidence for a production mechanism beyond editorial access. When training, collaborators and research time raise one another’s returns, funding them together can do more than expanding each separately.

A university supplies inputs used across departments: libraries, funding administration, computing and time for research. Advanced training, specialist feedback and collaborators are organised much more closely around disciplines. These specialised inputs help explain why a well-resourced university need not have a strong economics department. Section 7 places economics publication alongside overall university standing, while using author pools for the closer disciplinary comparison.

The model describes choices by researchers. It does not assign a social value to each topic. An observed-to-predicted topic ratio consequently measures a difference in research composition, not a welfare loss.

4 Data and measurement

I combine publication records, government speeches and a census of researchers, with THE rankings providing context on the wider universities. The same journal rules and fractional affiliation weights apply to South Africa and the comparator countries.

Leading-journal corpus. The companion study supplies indexed article records from 1990 to 2025 in nineteen economics journals: the conventional top five, selected general and field journals, and the principal development outlets.¹ Throughout the paper, “leading journals” and “frontier” refer to this fixed set. The top five are reported separately. The corpus holds 51,521 article records and 104,781 author slots. Historical *AER Papers and Proceedings* items are included in this inherited frame; regular top-five articles are distinguished when interpreting elite placement. Journal-subset and publication-type sensitivities appear in the appendix.

A title-and-abstract prefilter returned 430 possible articles about South Africa; detailed classification retained 246, and two corrupt bibliographic records were removed, leaving 244. I also

¹The journals are the *American Economic Review*, *Econometrica*, the *Journal of Political Economy*, the *Quarterly Journal of Economics*, the *Review of Economic Studies*, the *Review of Economics and Statistics*, the *Economic Journal*, the *Journal of the European Economic Association*, the *American Economic Journal: Applied Economics and Macroeconomics*, the *Journal of Development Economics*, *World Development*, the *Journal of African Economies*, the *Journal of Labor Economics*, the *Journal of Human Resources*, the *Journal of Public Economics*, the *RAND Journal of Economics*, the *Journal of International Economics* and the *Journal of Monetary Economics*. The *Journal of African Economies* is both a member of this set and a plausible outlet for this paper; the set was fixed in the companion census before this paper was designed.

hand-coded the main data source used in each article. Abstract coverage is incomplete in five Elsevier journals, with coverage of 7 to 15 per cent before 2010 and 31 to 66 per cent after (Appendix Table 16). Because the prefilter also reads titles and author affiliations, 22 per cent of the retained articles were detectable only through their abstract. Section 5 reports a coverage-corrected series alongside the observed one.

National publication corpus. The national sample starts with 15,214 OpenAlex works collected under the economics-field and `type:article` filters with at least one South African affiliation. OpenAlex covers African research output more widely than the main commercial databases, but its affiliation fields are less complete (Alonso-Álvarez and van Eck, 2025). The main sample requires the primary source to be typed as a journal with an ISSN, which retains 13,862 works, or 91.1 per cent. The excluded records include 21 non-journal-primary records linked by DOI to a journal record; these are part of the scope exclusion, not a further subtraction.

I allocate these publications to countries by their authors' affiliations. An article with m authors gives each author weight $1/m$, and an author's location is the first institution listed on the article. OpenAlex can reorder affiliation strings, so the original order was recovered from Crossref where necessary, and 93 ambiguous author positions in the nineteen-journal corpus were adjudicated by hand. The output series uses all journal-confirmed records. Topic analysis additionally requires an abstract: 12,587 of the collected works were submitted, 12,584 classified successfully, and one mismatched record was excluded after independent review. Journal restriction retains 11,412 classified records, of which 10,497 have positive South African first-affiliation weight, totalling 8545.2 article-equivalents. Appendix Table 6 reconciles these steps by period. The topic sample retains 84.4 per cent of national journal weight overall, rising from 61.4 per cent in 1990–94 to 89.0 per cent in 2020–24. Its topic shares therefore describe abstract-covered output.

Comparator publication corpus. The comparison sample is stratified by country and five-year period and covers 25 countries: Argentina, Bangladesh, Brazil, Chile, Colombia, Czechia, Egypt, Finland, Greece, India, Indonesia, Israel, Kazakhstan, Malaysia, Mexico, New Zealand, Poland, Portugal, Russia, South Korea, Thailand, Turkey, the United Arab Emirates, Uruguay and Vietnam. Cells with more than 1,500 articles are sampled at that size and carry expansion weights that recover the OpenAlex population; 133 of the 200 cells are full enumerations. Before fractional authorship is applied, the journal-and-ISSN rule retains 87.8 per cent of comparator expansion weight. Each comparator article receives its country-bin expansion factor multiplied by its domestic first-affiliation authorship fraction. The fraction is recovered from the ordered affiliation field in the OpenAlex deposit. For every author slot in the nineteen-journal corpus, it is reconstructed from the banked raw affiliation strings. Before using the reconstructed affiliations, I checked them against 816 adjudicated South African author positions: slot agreement was 94.5 per cent, errors were symmetric, and reconstructed South African leading-journal growth was 1.09 times the adjudicated figure (Appendix A). That audit covers South African positions, so it does not establish accuracy in each comparator country. Appendix Table 10 reports unresolved positions by

country and period and growth sensitivities to their allocation. The additional publisher corrections in Section 7 are applied throughout.

For each topic, an equally weighted fractional logit across 199 comparator country-period cells relates topic shares to log income per head, log population and five-year-period indicators. South Africa is excluded from estimation. Its predictions use South African covariates and each target portfolio’s own distribution of output over periods. The appendix specifies the model and tests alternative and matched-country benchmarks.

Policy speeches. The corpus contains 65 State of the Nation and Budget addresses from 1994 to 2026 that meet a fixed inclusion rule and contain classifiable text: 33 State of the Nation addresses with 5,141 paragraphs and 32 Budget speeches with 2,355 paragraphs. Paragraph probabilities are averaged within an address, and each address receives equal weight. Because Budget speeches are about fiscal policy by construction, the two series are reported separately.

Topic classification and validation. A fixed classifier assigns every article and policy paragraph probabilities over twelve topics: firms; labour; household income, poverty and inequality; fiscal policy and social protection; education; health; macroeconomics; trade; finance; institutions; race, ownership and affirmative action; and a residual class. An independent coder, blind to the model output, labelled a random sample of 400 articles with known inclusion probabilities. After four corrupt records were removed, primary-label agreement was 61.2 per cent. Agreement varies across topics. Finance has precision 0.89 and recall 0.89; the corresponding values for households are 0.40 and 0.55. Macroeconomics has precision 0.36 (Appendix Table 15).

The primary corrected topic estimand is the share that independent human primary labels would assign to each abstract-covered portfolio, under its article weights. A prediction-powered difference estimator combines all model probabilities with the design-weighted residuals from the validation sample (Angelopoulos et al., 2023; Ludwig, Mullainathan and Rambachan, 2025). Separately, a confusion-matrix sensitivity corrects both South African and comparator share vectors, refits the topic benchmarks and resamples the validation rows. This second procedure supplies the corrected ratios and classifier bands; it requires the classification matrix to transfer across countries and periods. The level estimator does not itself correct the comparator regression.

University rankings. The institutional comparison counts universities in the overall THE World University Rankings 2026 at the top-200, top-500 and top-1000 thresholds (Times Higher Education, 2025). It uses the same 26 countries as the publication comparison. Each ranked institution counts once; the entire 401–500 band belongs to the top 500. Reporter institutions have no rank and are excluded. A zero records no ranked institution within the cutoff, rather than an absence of universities or research. THE combines teaching, research environment, research quality, international outlook and industry measures, so this is a broad institutional measure (Times Higher Education, 2026). Its teaching- and research-reputation survey indicators carry 33 per cent of the weight. Its research-publication window is 2020–24. I therefore report a separate economics comparison for

those years alongside the full 1990–2025 record. The 2026 ranking describes contemporary standing and is not used as a historical control for the entire publication period.

Career census. Section 9 uses a census of leading-journal economists by country of education, built in Fourie and Wantchekon (2026) on the same nineteen journals and years. A person counts for a country if biographical evidence shows that they completed at least one stage of education there, whatever their current employer. Someone educated at the University of Cape Town and working abroad therefore counts for South Africa; employment at a South African university alone does not qualify. Candidates are drawn from journal authorship, affiliation histories and co-authorship; a language-model classifier reads each person’s curriculum vitae, university profile and biography, records evidence for each education stage and returns a verdict.

Two research assistants’ labels agreed too rarely to serve as a gold standard. A second language model therefore adjudicated a stratified probability sample of 608 records from the stored evidence. Of these, 316 home-education labels remain unknown and are excluded from the binary denominators; the unknown cases are unlikely to be random. Among resolved audit cases, design-weighted sensitivity of joint inclusion is 73 per cent and precision 95 per cent, indicating net omissions in those cases. Extending their classification residuals to unknown biographies requires an additional missingness assumption that the audit cannot verify. Figure 7 therefore reports machine-identified counts and doctoral-route bounds. The appendix states the assumptions behind the companion census’s corrected estimates. The fifteen-country set is the intersection of its pre-specified focus, Latin American, African and emerging groups with eligible national comparison units; it is held fixed here.

5 Research on South Africa and author affiliations

Research about South Africa became more common in the nineteen journals until the late 2000s, then declined. Figure 1, panel A, reports articles about the country per thousand articles in those journals, with 95 per cent Wilson intervals. Bins span five years except 2020–25, which spans six. The dashed line corrects for missing abstracts. The observed rate was 1.2 in 1990–94, 5.2 in 2000–04 and 7.1 at its 2005–09 peak. It fell to 4.5 in 2010–14 and has since remained near five. The coverage-corrected series, which adds the expected number of about-South-Africa articles among the Elsevier articles without abstracts, peaks at 8.2 and then declines by 34 per cent, against 30 per cent observed. Coverage improved after 2010, so the correction raises the peak more than the later bins. Under this coverage assumption, the adjustment strengthens the decline. The observed and high-coverage series provide comparisons that do not impute missing articles.

The fixed-window comparison gives weaker evidence of a change in top-five attention. Comparing 1990–2008 with 2009–2025 gives a top-five rate ratio of 0.59, with a 95 per cent conditional Poisson interval of 0.25–1.38 (Appendix Table 17). Excluding historical AER proceedings gives a ratio of 1.05 and an interval of 0.29–4.18 (Table 20). These small regular-article counts do not

establish a decline in top-five attention.

The observed top-five rate nevertheless falls from 3.6 per thousand in 2005–09 to 1.0 afterwards, a ratio of 0.27 when proceedings are included and 0.35 when they are removed. These are descriptive comparisons with a high point selected from the series; ordinary rate intervals would not account for that selection. Outside *World Development* and the *Journal of African Economies*, attention is approximately flat at two articles per thousand from 2000 onwards, with a post-2005–09 ratio of 0.92. The two development journals carry 64.8 per cent of articles about South Africa and drive the full-set peak and its aftermath.

The two development journals suit much of the research about South Africa. The *Journal of African Economies* specialises in African economic analysis, while *World Development* covers development problems across disciplines (Oxford University Press, 2026; Elsevier, 2026). They are natural outlets for country-specific research. Yet the topic labels do not account for the full authorship difference. Among the 244 articles about South Africa, domestic fractional authorship is 33.6 per cent in these two journals and 9.6 per cent in the other seventeen. Household topics make up 19.8 and 17.3 per cent of the two groups respectively. Within the household topic, the domestic authorship shares are 33.0 and 11.6 per cent; within labour, 33.2 and 7.9 per cent. These probability-weighted comparisons show that broad topic composition alone is insufficient to explain the concentration of domestic participation.

Domestic authors thus publish disproportionately in the development journals even within broad topics. Whether these journals are easier to enter requires submission and decision data: the published articles cannot show the same manuscript’s chances of acceptance at different outlets. The two journals account for 86.6 per cent of domestic fractional authorship *about* South Africa. Across all subjects published *from* South Africa in the nineteen-journal set, their share is 86.3 per cent. In both comparisons, most domestic authorship is in these two outlets. Counting all nineteen journals together obscures the much smaller domestic presence in the top five.

Household surveys are the main data source in nearly half the articles. Figure 2 groups articles by their main data source, hand-coded from abstracts and full texts. Bubble area measures article counts and colour the South African first-affiliation authorship share. The dates mark the 1993 living-standards survey (PSLSD), the KwaZulu-Natal Income Dynamics Study (KIDS), the National Income Dynamics Study (NIDS), and the 2014 Treasury–UNU-WIDER tax-data call. Household surveys appear in 110 articles, or 45.1 per cent, and their count rises across periods. Firm and tax data appear in 13 articles. The firm-data class includes surveys, customs and tax records; it should not be read as a count of applications of the later SARS–NT panel.

Figure 1, panel B, allocates fractional authorship to South African-based authors, the South African-educated diaspora and all other authors. Foreign-based authors hold at least 57 per cent in every period and 78 per cent in 2000–04. The domestic share fell from 21 per cent in 1990–94 to 14 per cent in 1995–99, then rose to 33 per cent in 2015–19; it is 29 per cent in 2020–25. Thus the declining article rate after 2009 did not coincide with a declining domestic authorship share. Foreign authors dominate development research on other countries too: in leading development journals,

most articles over 1990–2019 have no author based in a developing country (Amarante and Zurbrigg, 2022), although North–South coauthorship has risen (Amarante and Zurbrigg, 2024). Most of that domestic gain occurred in the two development journals. Across all topics in the nineteen-journal corpus, 221 authors published with a South African first affiliation. They occupy 5 author positions in the *American Economic Review* and none in the other four top-five journals. Two positions are in regular articles and three in historical *Papers and Proceedings* contributions (Appendix Table 19).

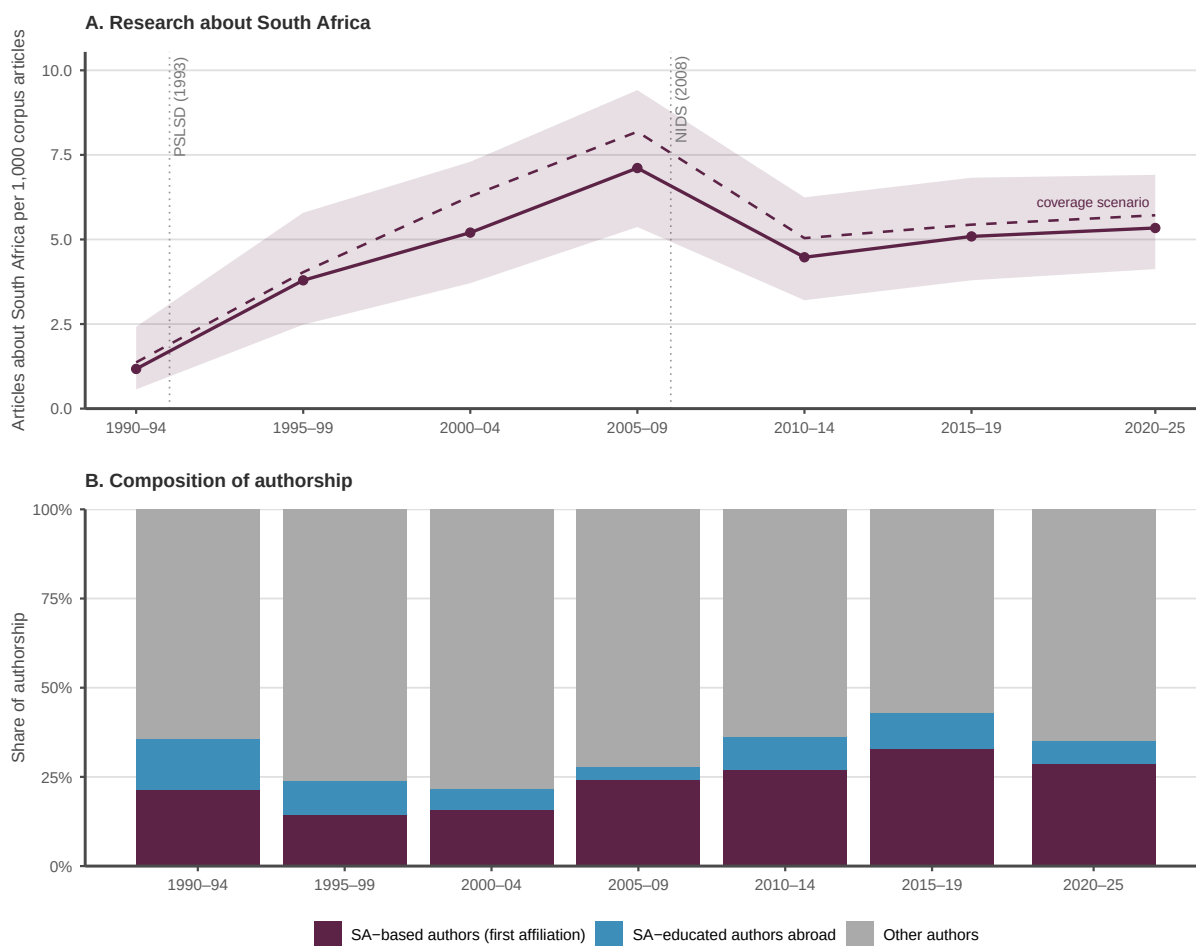


Figure 1: Research on South Africa: publication rates and authorship.

Notes: Panel A: articles about South Africa per thousand articles in the nineteen leading journals, with 95 per cent Wilson intervals; bins span five years except 2020–25. The dashed line adds the expected number of about-South-Africa articles among Elsevier articles without abstracts. Panel B: fractional authorship of those articles by South African-based authors (first affiliation), South African-educated authors abroad and all other authors.

6 Topic composition

Domestic researchers devote the largest share of their work to finance; leading-journal research about South Africa concentrates on households. Figure 3 divides each topic’s observed share by

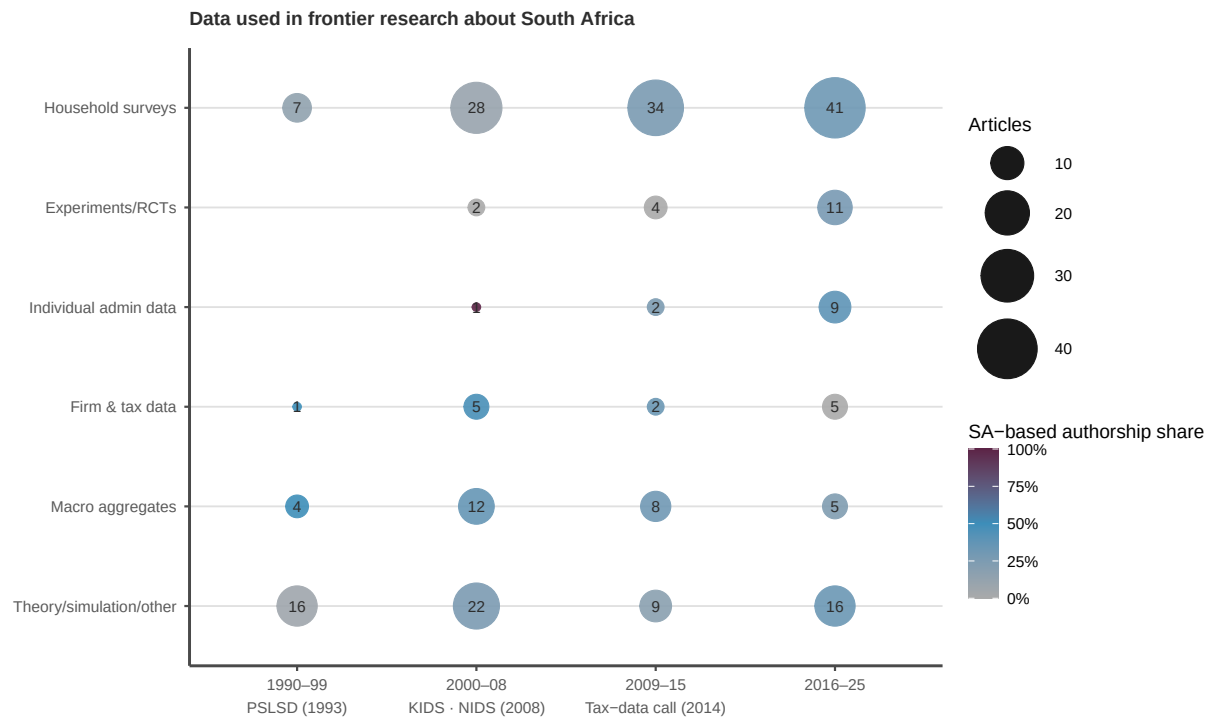


Figure 2: Data sources in leading-journal research on South Africa.

Notes: Articles about South Africa grouped by their main data source, hand-coded from abstracts and full texts. Bubble area is the number of articles; colour is the South African first-affiliation share of fractional authorship. Dates mark the 1993 PSLSD, KIDS, NIDS and the 2014 Treasury–UNU-WIDER tax-data call.

the fractional-logit prediction for a country with South Africa’s income and population. Ratios above one indicate more research on a topic than predicted; ratios below one indicate less. The plot uses a log scale, with domestic production on the left and leading-journal work about South Africa on the right. Grey points divide the pooled comparator share by the same prediction. Two intervals accompany each point: the envelope of a country-block and an author-block bootstrap, and a paler band for classifier error. The hollow marker is the point estimate after the symmetric confusion-matrix correction. The bands show sampling and classification uncertainty separately, rather than a combined confidence interval.

Figure 4 shows how domestic research differs from the comparators within and across topics. Following the visual approach of Harris (2026), the map places articles with similar content near one another. Rose cells indicate a larger share of domestic research than of comparator research in that region; blue cells indicate a smaller share. Its benchmark is the pooled comparator portfolio, whereas Figure 3 adjusts for income, population and period. Labels mark topic centres; cells with fewer than 30 comparator records are neutral. Appendix A explains the map’s construction and the interval and axis conventions.

Leading-journal work about South Africa is concentrated on households. The topic accounts for 18.9 per cent of that work against a predicted 4.7 per cent, a ratio of 4.02. The correction raises the ratio to 7.73, with a classifier band from 1.92 to 13.72. Household concentration also survives the alternative benchmarks in Appendix Table 7: its uncorrected ratio ranges from 4.02 to 5.00 across specifications and from 3.92 to 4.22 when comparator countries are omitted one at a time. Labour and education are also above prediction. Race, ownership and affirmative action has a small level but a large ratio because its predicted share is close to zero, consistent with the concentration of race-related economics in particular settings (Advani et al., 2026). Finance, health, firms and the residual class are below prediction.

Finance is the largest domestic topic. Its model share is 20.5 per cent and its primary corrected share is 27.4 per cent, with a 95 per cent interval of 20.8–33.3 (Appendix Table 8). Its uncorrected ratio ranges from 1.25 to 1.46 across the benchmark specifications; the confusion-matrix correction gives 1.61, with a classifier band from 1.43 to 1.88. The domestic household share is much smaller: 2.4 per cent after the primary correction, with an interval from -2.2 to 7.6. Its ratio to the comparator prediction is closer to one and more sensitive to the benchmark than household research about South Africa.

Firm-topic shares are below prediction in both portfolios under every benchmark tested. The domestic model ratio ranges from 0.54 to 0.64, and the about-South-Africa ratio from 0.57 to 0.64. The size of the shortfall is less well measured than these stable model comparisons might suggest. The primary corrected domestic firm share is 4.4 per cent, with an interval from -0.2 to 9.3. The constrained matrix inversion often places the about-South-Africa firm share at zero, so its classifier band does not give a reliable measure of precision. The evidence is consistent with a relatively small firm portfolio; it does not precisely establish the size of its difference from the benchmark. An earlier review of education economics also found a portfolio tied closely to available surveys

(Gustafsson and Mabogoane, 2012).

Table 1 reports topic shares in per cent for the domestic, about-South-Africa and pooled comparator portfolios alongside government speeches. Its policy columns show all 65 addresses and the separate State of the Nation and Budget series, with equal weight per address. All columns use the same twelve-topic instrument; BEE denotes race, ownership and affirmative action. The two speech series differ by construction: fiscal policy takes 42.7 per cent of Budget speeches and 7.2 per cent of State of the Nation addresses, while institutions take 28.6 per cent of the latter. The classified research and speech portfolios differ. Institutions are 28.6 per cent of State of the Nation attention and about 11 per cent of domestic research. Finance is 0.8 per cent of State of the Nation attention and 20.5 per cent of domestic output. Households take about 4 per cent of policy attention in both series but 18.9 per cent of leading-journal attention. These speech shares describe political attention, rather than demand for commissioned research. They also have a different validation status: the classifier was checked against human labels for articles, but not speeches. These cross-genre comparisons are exploratory: using one instrument does not establish equal classification accuracy in political and academic text.

The comparisons identify differences in attention, rather than an optimal allocation of research. Household research can be valuable, and finance matters in a country with deep capital markets. The relatively small measured firm portfolio raises a more specific question: did the subsequent availability of linked tax records bring more domestic research on firms? Section 8 examines that change.

Table 1: Topic shares in research and policy speeches.

Topic	All addresses	SONA	Budget	SA- produced	About-SA frontier	Comparators
Fiscal	24.7%	7.2%	42.7%	5.7%	8.4%	9.5%
Institutions	20.7%	28.6%	12.6%	10.8%	13.4%	12.0%
Macro	15.7%	14.9%	16.5%	17.4%	11.7%	16.2%
Other	10.1%	15.7%	4.2%	7.5%	1.6%	11.4%
Labour	5.2%	6.0%	4.3%	4.6%	13.4%	4.6%
Education	4.5%	5.3%	3.7%	1.9%	5.7%	2.6%
Firms	4.5%	5.5%	3.4%	7.7%	7.4%	12.3%
Households	4.2%	4.6%	3.7%	5.6%	18.9%	4.8%
Trade	3.0%	4.1%	1.9%	6.5%	6.2%	6.7%
Health	2.7%	2.8%	2.7%	11.1%	4.4%	5.3%
BEE	2.7%	4.5%	0.9%	0.5%	5.0%	0.1%
Finance	2.1%	0.8%	3.3%	20.5%	3.8%	14.5%
	65 addresses	33 addresses	32 addresses			
	(7,496	(5,141	(2,355	10,497	244	126,832
N	paragraphs)	paragraphs)	paragraphs)	articles	articles	articles

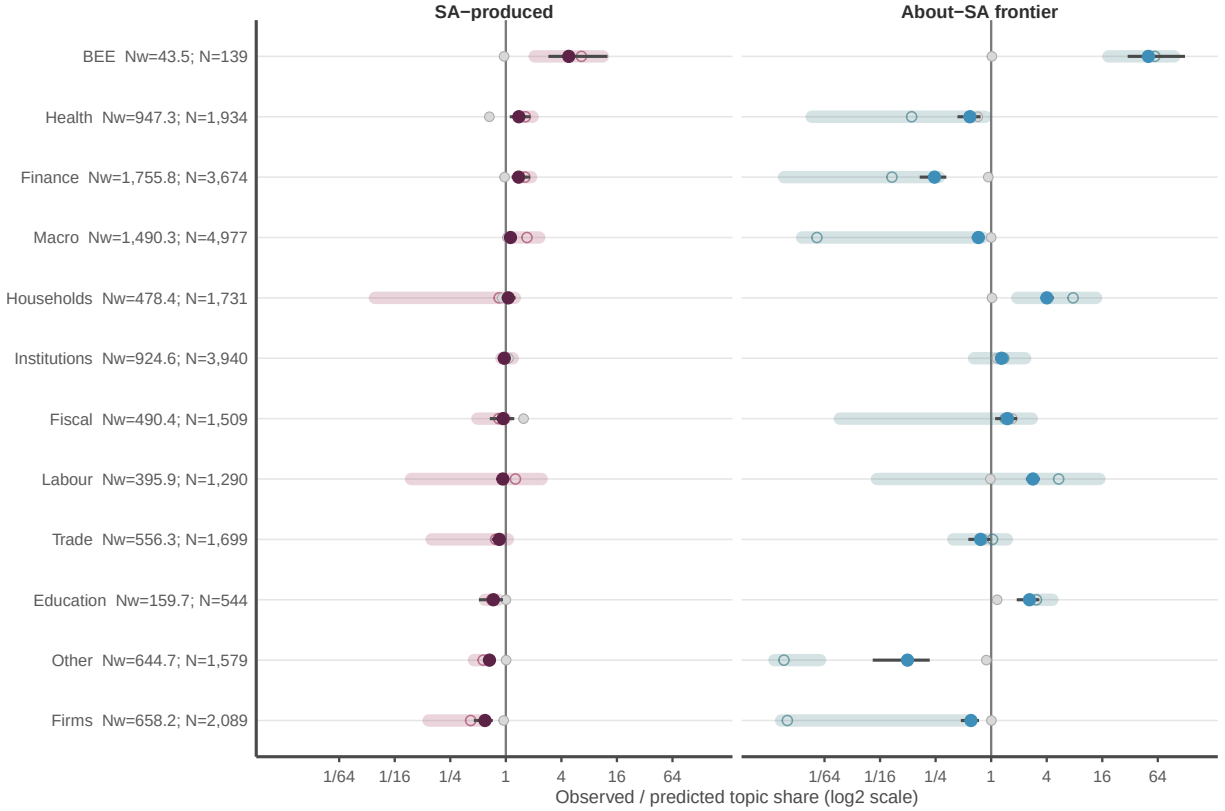


Figure 3: Observed and predicted topic shares.

Notes: Observed topic share divided by the fractional-logit prediction for a country with South Africa's income, population and period mix (log scale). Left: domestic output (South African first affiliations); right: leading-journal research about South Africa. Grey points: pooled comparators. Dark intervals: envelope of country-block and author-block bootstraps; pale bands: classifier error. Hollow markers: symmetric confusion-matrix correction.

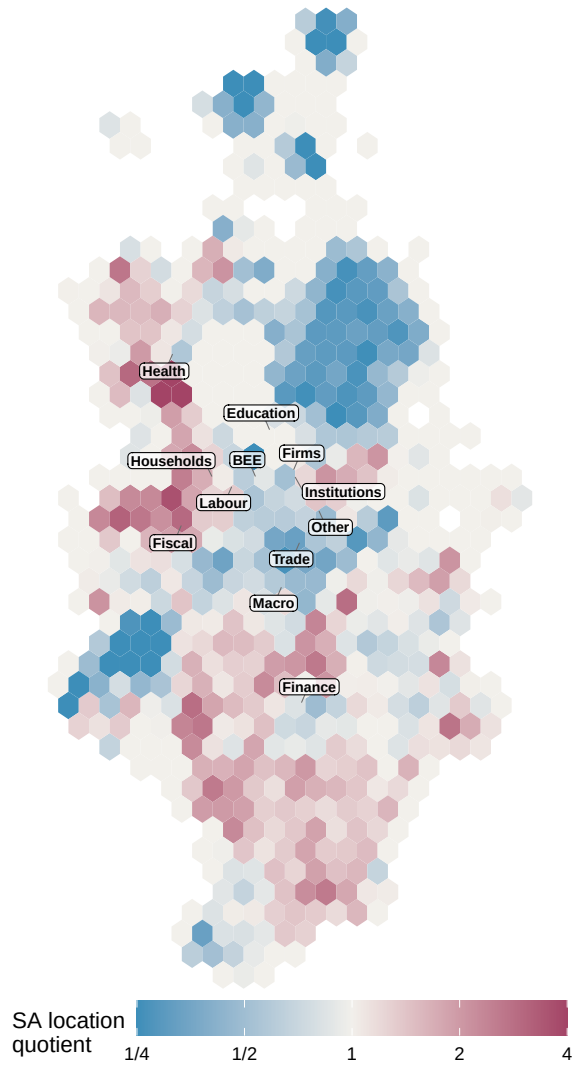


Figure 4: Topic map of South African research.

Notes: Articles are placed by the similarity of their text embeddings. Rose cells hold a larger share of domestic than of pooled comparator research; blue cells a smaller share. Cells with fewer than 30 comparator records are neutral. Labels mark topic centres. Appendix A describes the construction.

7 Publication growth, journal placement and citation impact

7.1 Publication by journal tier

Table 2 ranks all 26 countries by the share of their nineteen-journal output published in the conventional top five. It reports article counts for three nested sets: the top five, a specified ten and the full nineteen. The ten add the *Review of Economics and Statistics*, *Economic Journal*, *Journal of the European Economic Association*, *AEJ: Applied Economics* and *AEJ: Macroeconomics*. The ten-journal set tests whether the result depends on the narrow top-five definition; there is no universally agreed ranking of journals six to ten. Historical AER proceedings are excluded from all three journal sets. Entries count distinct articles with at least one domestic first affiliation; parentheses give fractional article-equivalents. All located author positions contribute, including those without author IDs. The top-five percentage divides regular-top-five article-equivalents by the nineteen-journal total; the final column gives the number of THE top-500 universities in 2026. The publication columns cover 1990–2025. The concentration analysis in Table 3 requires IDs.

South Africa’s 2 regular top-five articles contain one domestic author each, equivalent to 0.58 fractionally allocated articles over 36 publication years. There are no domestic first affiliations in *Econometrica*, the *Journal of Political Economy*, the *Quarterly Journal of Economics* or the *Review of Economic Studies*. The ten-journal comparison raises the total to 11 articles, or 4.97 article-equivalents. All four Latin American comparators exceed it. The distinction persists in 2015–24: South Africa has eight ten-journal articles, against 62 in Chile, 51 in Brazil, 20 in Colombia and 16 in Argentina. Its two regular top-five articles compare with 17, 20, five and seven respectively. The result therefore extends beyond one narrow journal definition and the early years of the sample.

South Africa’s regular top-five articles account for 0.5 per cent of its nineteen-journal fractional output, against 2.6 per cent in Colombia, 8.4 in Chile, 9.6 in Brazil and 11.1 in Argentina. That denominator includes a geographically specific African journal and *World Development*, so the share also reflects the country’s outlet mix. Removing JAE raises South Africa’s share to 0.7 per cent; removing both development journals raises it to 3.8 per cent; using only the ten journals gives 11.7 per cent. Its full-period rank remains 24th of 26 in all four comparisons, above Egypt and Malaysia (Appendix Table 5). The level of the share is sensitive to the basket, but the long-period ordering is not.

South Africa has more highly ranked universities than several countries that publish far more top-five economics articles. Cape Town ranks joint 164th globally, Stellenbosch and Witwatersrand lie in the 301–350 band, and Johannesburg in 351–400. South Africa consequently has 4 universities in the global top 500, more than Brazil’s two, Chile’s one and Czechia’s one (Times Higher Education, 2025). Those countries have 39, 30 and 21 regular top-five economics articles respectively, against South Africa’s 2. New Zealand has the same number of top-500 universities as South Africa, but 14 regular top-five articles and a top-five share of 6.3 per cent. Across the full set, 16 countries combine fewer top-500 universities with more regular top-five articles than South Africa.

The university comparison locates the economics result within a wider institutional setting.

It does not provide an expected-output model: university-wide rankings combine disciplines, and a count above a threshold contains no information about economics faculty size or research time. Reputation, history and location also raise the standing of leading national universities (Selten et al., 2020), so overall rank can diverge from economics research output. The author-pool comparison below is closer to the question of disciplinary capacity.

The absolute counts remain small in 2020–24, the publication window used by THE (Appendix Table 13). It has 2 regular top-five articles and 7 in the ten-journal set, against eight and 32 in Chile, eleven and 30 in Brazil, four and 17 in Czechia, and four and 15 in Colombia. Its top-five share is 1.9 per cent, ranking 22nd of 26 with all nineteen journals. Excluding JAE and *World Development* raises it to 9.1 per cent and 16th; the ten-journal denominator gives 19th. The recent rank is therefore sensitive to outlet composition. Argentina has two and six articles respectively in this window, so the absolute gap does not persist against every peer.

The appendix also changes the institutional cutoff to the top 200 and top 1000. South Africa has 1 and 8 institutions at these thresholds; the contrasts with Chile and Czechia persist, while Brazil matches South Africa’s top-1000 count. These cutoffs describe the same institutional contrast; changing them does not turn overall university rank into an economics-capacity measure.

These are publication-time institutional affiliations, which include researchers outside economics departments; they do not identify nationality. Affiliation order matters: checking the published records removes two apparent South African ten-journal contributions whose authors list Monash and UC Davis first and South African universities second. The corrections are propagated through the domestic placement series and author counts.

7.2 Growth and citation impact

Domestic publication and citation impact have grown substantially, while publication in the nineteen leading journals has grown more slowly (Figure 5). Each series is a trailing three-year average indexed to its own 2000–04 mean. Only complete three-year windows are shown, so all series use the same calendar years at a common endpoint. Journal output is fractional South African first-affiliation weight in the journal-confirmed national corpus. Leading-journal output applies the same rule to the nineteen journals. Top-decile impact counts national-corpus articles whose citations meet or exceed the ninetieth percentile of a 1,000-article world economics sample from the same publication year; the series ends in 2020 because later rankings are immature. Citation counts and thresholds share an extraction date. In 2025 the journal-output index is 830 and the leading-journal index 259. By 2020 the top-decile index had reached 741, above the 509 for output. Appendix Figure 10 reports the underlying annual levels.

Faster growth in total output than in leading-journal publication could reflect publication rewards, wider database coverage or publication in more journals. Figure 6 tests whether South Africa’s growth pattern differs from its peers, using growth in journal-confirmed output Q and nineteen-journal placement F for South Africa and the 25 comparators, both measured in fractional first-affiliation article-equivalents. Appendix Table 18 also reports growth in top-decile citation im-

Table 2: Journal placement and university rankings.

Country	Regular top five	Ten journals	Nineteen journals	Top-five share (%)	THE top 500 (2026)
Russia	68 (36.5)	137 (72.1)	188 (102.3)	35.7	3
Israel	169 (100.7)	318 (198.6)	545 (348.2)	28.9	3
Czechia	21 (10.6)	57 (29.9)	95 (52.3)	20.3	1
Turkey	29 (13.6)	42 (18.6)	127 (71.4)	19.1	4
South Korea	55 (27.8)	104 (52.9)	291 (165.6)	16.8	14
Portugal	23 (12.0)	66 (34.7)	140 (79.8)	15.0	3
Kazakhstan	3 (0.8)	5 (1.8)	12 (5.0)	15.0	1
Greece	7 (5.0)	19 (11.0)	54 (34.0)	14.7	1
Poland	5 (1.7)	13 (4.4)	27 (15.0)	11.2	0
Argentina	21 (10.2)	42 (21.5)	156 (91.8)	11.1	0
India	71 (34.6)	150 (74.7)	557 (341.2)	10.1	4
United Arab Emirates	5 (2.7)	14 (7.1)	51 (26.7)	10.0	7
Brazil	39 (19.4)	85 (44.6)	321 (202.5)	9.6	2
Chile	30 (15.6)	98 (52.0)	305 (184.5)	8.4	1
Mexico	27 (11.6)	71 (32.2)	252 (154.5)	7.5	0
Uruguay	3 (1.2)	11 (4.5)	30 (16.8)	7.3	0
Finland	16 (8.3)	64 (38.3)	202 (128.8)	6.5	9
New Zealand	14 (6.3)	51 (28.8)	165 (100.8)	6.3	4
Thailand	3 (0.8)	6 (1.8)	46 (25.2)	3.3	0
Bangladesh	5 (1.2)	20 (7.3)	87 (43.5)	2.7	0
Colombia	8 (2.9)	27 (11.3)	192 (110.2)	2.6	0
Indonesia	4 (1.2)	9 (2.4)	102 (53.2)	2.3	0
Vietnam	1 (0.3)	3 (1.2)	50 (22.9)	1.5	0
South Africa	2 (0.6)	11 (5.0)	194 (118.7)	0.5	4
Egypt	0 (0.0)	2 (0.6)	31 (17.8)	0.0	0
Malaysia	0 (0.0)	4 (1.6)	24 (11.6)	0.0	7

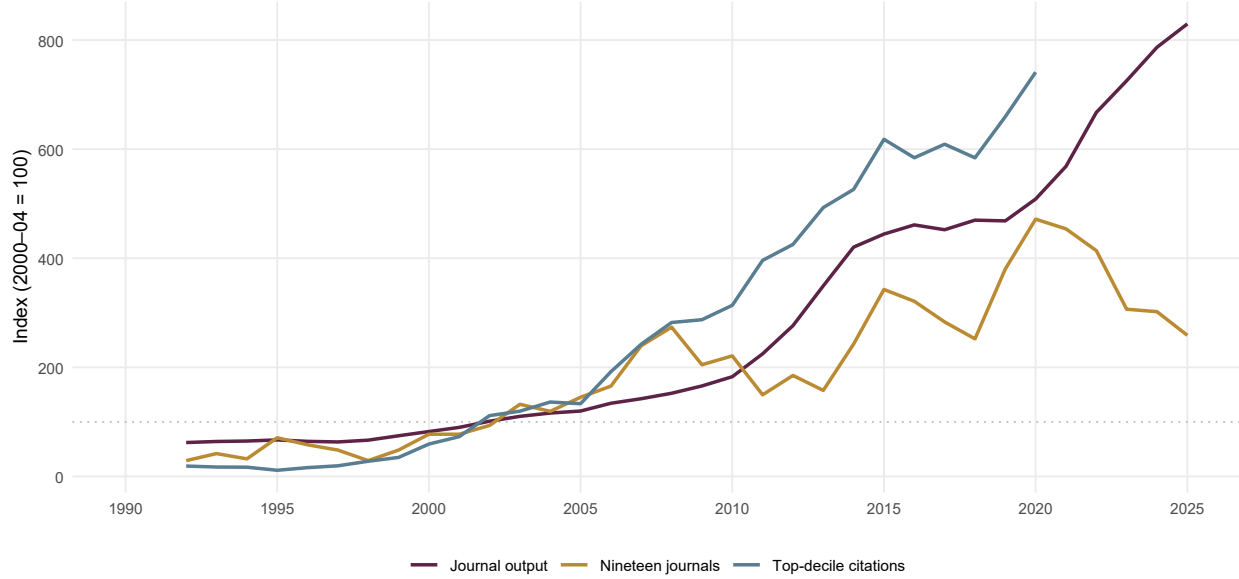


Figure 5: South African publication, journal placement and citation impact.

Notes: Trailing three-year averages indexed to each series' 2000–04 mean; only complete windows are shown. Journal output: fractional South African first-affiliation weight in the journal-confirmed national corpus. Leading journals: the same rule in the nineteen journals. Top-decade impact: articles at or above the ninetieth citation percentile of a world economics sample from the same year, to 2020.

pact T under the same authorship rule.

For comparability, F uses the common reconstruction of affiliation order in every country. The dedicated South African series in Figure 5 uses adjudicated affiliations; its corresponding placement growth is 2.2 times, compared with 2.4 under the common reconstruction. Growth is measured from 2000–09 to 2015–24 for Q and F and to 2011–20 for T . The statistic is $g_c = \log(F \text{ growth}) - \log(Q \text{ growth})$, leading-journal growth net of output growth. A negative value means total output grew faster than publication in the nineteen journals.

Countries whose base placement is below five article-equivalents are shown hollow and excluded from the fitted line; there are 7 of them. The dotted line is the 45-degree line and the solid line is the comparator fit. South Africa's error bars are marginal 95 per cent intervals for each axis from a joint bootstrap of publication IDs; they are not a joint confidence region. The underlying OpenAlex records were downloaded on 25–27 August 2026.

The median comparator's output grew 6.6 times and its placement 1.7 times. South Africa's output grew 4.5 times and its placement 2.4 times, so g is -0.63, with a joint article-bootstrap interval from -1.05 to -0.21. That ranks 6 of 26 countries in the unrestricted source records. South Africa lies +0.47 log points above the comparator line, within its wide prediction interval. Alternative affiliation rules and leading-journal sets give similar ranks within that source frame; Appendix A reports the estimates. On a comparable citation window, top-decade impact grew 3.2 times and output 3.4 times. Citation impact thus broadly kept pace with total output, despite limited publication in the highest-tier journals.

Indonesia’s exceptionally large output increase partly reflects the journals represented in its later records. Its output grows 70.94 times from 619 to 43,932 fractional article-equivalents, compared with 1.56 times in the nineteen journals. The large output ratio in Figure 6 reproduces from the frozen samples and their population weights. However, 96.7 per cent of its later output is in journal sources absent from its 2000–09 records. Retaining the source IDs observed in that base reduces output growth to 2.36 times. The later records span a large range of local economics, accounting and management outlets. This decomposition combines changes in publication venues and database coverage; it is not evidence of a seventyfold expansion in internationally competitive economics research. Removing Indonesia leaves South Africa sixth of 25 countries. Appendix A documents the source audit.

Changing source representation matters beyond Indonesia. Applying the same rule to every country, 71.6 per cent of South Africa’s later output comes from source IDs absent from its base-period records. Output in base-period sources grows 1.27 times, compared with 2.4 times in the nineteen journals. The resulting net growth statistic is 0.62, and South Africa ranks 10th of 26 (Appendix Table 9). Placement now outgrows base-source output in 18 of the 26 countries, whereas it outgrows unrestricted output only in Finland. The broad volume-placement gap is therefore sensitive to the source frame.

Holding source IDs fixed removes entry into journals as well as changes in index coverage. It can also miss base-period sources in sampled country cells. Neither series isolates changes in research productivity, but their comparison shows why the original sixth-place rank should not carry the interpretation of South Africa’s performance. The more direct result remains its small five- and ten-journal presence and its fewer repeat authors.

Growth has not continued throughout the recent period. South Africa’s placement index in Figure 5 falls from 472 in 2020 to 259 in 2025, and Table 3 records no author with three or more articles in 2020–25. The counts are small and the final year is incomplete, so a sustained decline cannot yet be established.

Fewer South African authors return to these journals than their Chilean counterparts. Table 3 reports fractional output by decade and pooled, using adjudicated domestic affiliations for South Africa and reconstructed affiliations for Chile, Colombia and Czechia. Over 36 publication years, 221 South African-based authors produced 119 article-equivalents in the nineteen journals, on 196 articles. The top ten authors hold 15.5 per cent of that output, the inverse Herfindahl index counts 133 effective authors, and 20 authors have three or more leading-journal articles. Chile has 253 authors, 186 article-equivalents, 38 authors with three or more articles and 47 top-five author positions under the same broad definition, including historical AER proceedings; Czechia, with 82 authors against South Africa’s 221, has 32.

Holding author numbers equal preserves the gap. I draw 221 authors at random from Chile’s pool 999 times. The draws give a median of 163 article-equivalents, against South Africa’s 119, and 29 to 37 authors with three or more articles, against 20. They yield 35 to 46 top-five positions, against South Africa’s 5, even with proceedings included. The top ten authors account for 16.6 to

19.0 per cent of output in the Chilean draws, above South Africa’s 15.5 per cent. South Africa’s output is therefore less concentrated among a few prolific authors: its authors publish less each, fewer return to the journal set, and they are almost absent from the top tier. Section 10 examines training, research connections, career choices and publication incentives as possible explanations.

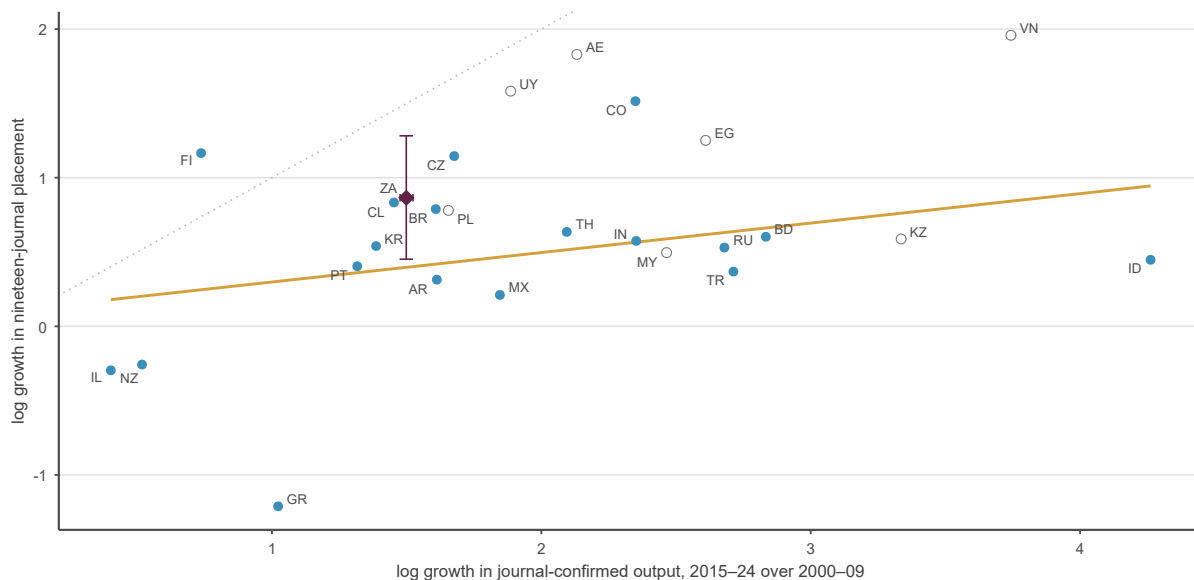


Figure 6: Publication growth and journal placement across countries.

Notes: Growth from 2000–09 to 2015–24 in journal-confirmed output (Q) and nineteen-journal placement (F), in fractional first-affiliation article-equivalents. Hollow points: base placement below five article-equivalents, excluded from the fitted line. Dotted: 45-degree line; solid: comparator fit. South African error bars are marginal 95 per cent intervals from a joint bootstrap of publication IDs.

8 Administrative data and firm research

Administrative data have expanded the questions South African economists can answer. Linked company-tax, payroll and customs records make it possible to study firm productivity, job creation and exporting together. Published applications include manufacturing markups (Fedderke, Obikili and Vieg, 2018) and the employment and wage premiums of exporters (Bezuidenhout, Matthee and Rankin, 2019).

The domestic share of firm research changed little relative to peers after the mid-2010s. Across the full sample, its model share is 0.60 of the comparator prediction, with the classification uncertainty described in Section 6. Comparing 2008–15 with 2016–24, its change relative to the equal-country comparator mean is -0.09 percentage points. Appendix Figure 11 shows the underlying shares and their relative movements, and Table 21 reports the comparisons. Access developed from the 2014 research call through later projects, and the comparator countries are not a verified untreated group, so 2016 is a comparison cutoff rather than an identified access shock.

A payroll-data study may be classified as labour or inequality research, while a firm study

Table 3: Concentration of domestic nineteen-journal output.

Country	Period	Authors	Article- eq.	Top 5 (%)	Top 10 (%)	Top 20 (%)	Effective authors	≥ 2 articles	≥ 3 articles
South Africa	1990–2025	221	119	9	16	27	133	49	20
	1990–99	12	8	59	91	100	10	0	0
	2000–09	49	27	24	42	63	35	8	1
	2010–19	100	49	13	24	42	69	20	3
	2020–25	97	35	14	24	39	70	9	0
Chile	1990–2025	253	186	10	16	27	138	77	38
	1990–99	27	22	37	61	88	20	8	2
	2000–09	58	39	20	33	59	43	15	4
	2010–19	115	68	16	27	42	69	28	10
	2020–25	103	58	16	27	45	66	25	9
Colombia	1990–2025	210	112	12	22	36	101	45	17
	1990–99	8	8	82	100	100	5	2	1
	2000–09	30	16	41	57	86	17	4	1
	2010–19	84	39	21	36	51	49	12	3
	2020–25	116	48	17	27	43	68	23	5
Czechia	1990–2025	82	55	26	40	56	41	19	12
	1990–99	4	2	100	100	100	4	0	0
	2000–09	12	9	64	93	100	10	4	0
	2010–19	32	23	42	60	82	17	9	6
	2020–25	46	21	22	41	65	36	11	4

may use survey data. The replication inventory therefore distinguishes original applications of linked firm records, individual tax data and panel documentation. Checking published studies also identifies applications missed by a title-and-abstract search, which makes an annual uptake rate from that search inappropriate. The near-zero change in topic share therefore does not imply that the new records went unused. For universities and data providers, the practical question is how to combine access with the training, research time and collaboration needed to use the records.

9 Researcher counts and doctoral training

South Africa has educated researchers who now publish abroad as well as at home. The career census in Section 4 counts them by country of education, rather than by the affiliations on their articles. Figure 7 compares South Africa with fourteen peers in the career-census comparison set.

Panel (a) shows machine-identified authors per million residents on a log scale. South Africa has 111 identified authors, or 1.84 per million, below Uruguay, Chile, Czechia, Argentina, Colombia and Turkey and above the other eight peers. These are counts within the machine-identified frame, not estimates corrected for unknown biographies. They include South African-educated authors now abroad and cover all nineteen journals. The domestic top-five comparison has a different location rule and journal scope.

Counting articles by country of education addresses a different question from the domestic count: how much leading research the country’s educated economists produce, wherever they work. Table 4 links the identified authors to their articles. An article counts once for a country if at least one author was educated there; home-based articles have such an author with a first affiliation in the country. The 111 South African-educated authors have six regular top-five articles, 39 in the ten-journal set and 183 in the nineteen journals. Of these top-five articles, two are home-based. Chilean-educated authors have 108 regular top-five articles and Colombian-educated authors 42; per million residents, the rates are 5.5 in Chile, 0.8 in Colombia and 0.10 in South Africa. South Africa ranks 13th of the fifteen countries by this count, above only Malaysia and Thailand. In every Latin American peer, as in South Africa, most top-five articles by the country’s educated economists are written from abroad. The difference lies in how many such articles South Africa’s educated economists produce.

These counts are lower bounds. The career census draws its candidates from authors who have held an affiliation in the country or coauthored with someone who has, so economists educated there who never did either are missed; this rule is common to all countries. The South African-educated authors abroad do not concentrate on their home country: 19 per cent of their 104 nineteen-journal articles written from abroad are about South Africa, against 68 per cent of those written from South Africa. The diaspora’s small leading-journal output, rather than a focus on South Africa, limits its contribution to the country’s count.

Panel (b) shows foreign doctoral training among the same identified authors. Of 102 South African cases with a known PhD country, 48 trained abroad, or 47.1 per cent. Nine doctoral coun-

tries are unknown. Assigning all nine to domestic training or all nine to foreign training gives a full-frame range of 43.2–51.4 per cent. The corresponding ranges are 84.2–89.5 per cent in Uruguay, 89.6–93.1 in Chile, 89.3–92.6 in Argentina and 85.5–96.1 in Colombia (Appendix Table 11). Thus unknown doctoral locations within the identified frame do not account for these route differences. Czechia’s range is 27.7–33.8 per cent, showing that the association is not uniform across peers. Economics doctoral students in the United States, including those from abroad, come disproportionately from families with highly educated parents (Stansbury and Schultz, 2023), so access to this route is itself unequal.

The points use known doctoral countries; horizontal lines assign every unknown route to one category or the other. These are missing-data bounds, not sampling confidence intervals. They hold the identified author frame fixed and do not correct biographies that the census omitted or misclassified. The comparison describes routes among eventual leading-journal authors; it cannot measure the probability that a domestic or foreign PhD reaches those journals.

Chile has more eventual leading-journal authors in each of three PhD cohorts. Among economists who completed their undergraduate education at home, later published in the journal set and have a known PhD year, South Africa has 5 doctorates from the 1980s, 7 from the 1990s and 16 from the 2000s; Chile has 12, 24 and 46. These are PhD-completion cohorts selected on eventual publication, so they say nothing about people who trained and did not reach the journal set, and early cohorts are thinned by survivorship. The comparison identifies where eventual authors trained, not where they subsequently worked.

Table 4: Leading-journal articles by authors’ country of education, 1990–2025.

Country	Identified authors	Regular top five	Top five, home-based	Ten journals	Ten, home-based	Nineteen journals	Top five per million
India	518	379	26	670	47	1560	0.26
Turkey	166	245	20	375	33	660	2.88
Argentina	215	233	17	365	36	767	5.04
Brazil	281	117	30	241	70	484	0.54
Chile	202	108	26	214	73	517	5.51
Poland	64	65	3	91	6	143	1.77
Mexico	84	57	7	77	11	205	0.44
Colombia	152	42	7	103	20	294	0.80
Uruguay	38	26	2	53	9	109	7.65
Indonesia	29	25	4	36	6	62	0.09
Czechia	65	18	13	54	34	95	1.71
Egypt	16	9	0	15	2	40	0.08
South Africa	111	6	2	39	11	183	0.10
Malaysia	8	5	0	12	0	28	0.15
Thailand	12	2	2	6	2	13	0.03

Notes: Articles with at least one author identified by the career census as educated in the country, wherever the author worked. Home-based articles have such an author with a first affiliation in the country. Regular top five excludes historical AER proceedings; the ten journals are defined in Table 2. Identified authors are machine classified and uncorrected for unknown biographies. Per-million rates use the census populations.

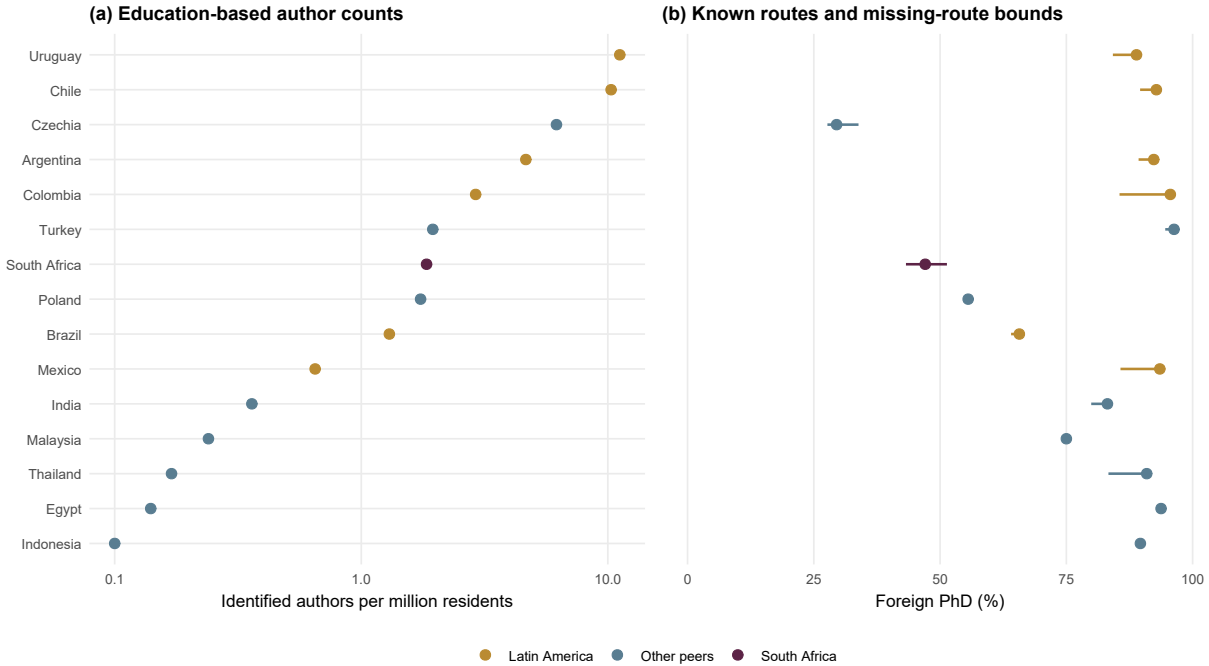


Figure 7: Researcher counts and doctoral training across countries.

Notes: Panel (a): machine-identified career-census authors per million residents (log scale), counted by country of education and uncorrected for unknown biographies. Panel (b): share of identified authors with a known PhD country who trained abroad; horizontal lines assign all unknown doctoral countries to one category or the other.

10 Potential mechanisms

Economists need specialised training, collaborators and time to develop work for the top journals. A university's broader standing does not ensure that its economics department supplies these conditions, or that promising graduates choose research careers. The comparisons above suggest several possible explanations for South Africa's small domestic presence at the top.

Doctoral training and research networks. South African authors have fewer recorded ties to established top-five authors. For each domestic author position in 2015–24, I ask whether the author had previously coauthored, within the nineteen-journal corpus, with someone who had already published a regular top-five article. Both the collaboration and the partner's top-five publication must precede the current publication year. The share is 7.0 per cent for South Africa, compared with 14.3 per cent in Colombia, 22.4 in Brazil, 28.9 in Chile and 34.4 in Argentina (Appendix Table 14). Giving each author equal weight instead of each author position preserves the ordering: 5.6 per cent in South Africa, against 9.1–27.9 per cent in those four peers. Excluding JAE and *World Development* leaves two connected positions out of 32 in South Africa (6.2 per cent), against 23.9–45.8 per cent in the four peers. In the ten-journal set South Africa has none among fourteen positions. Weighting the pooled four peers to South Africa's journal-by-five-year-

period distribution gives a connection rate of 14.0 per cent, against South Africa’s 7.0 per cent; all 171 South African positions have peer support in their journal-period cell. Outlet composition accounts for part of the gap, but does not remove it (Appendix Table 12).

These ties are measured among people already publishing in the nineteen journals; unrecorded working-paper collaborations, seminars and mentorship are outside the measure. The South African positions have a mean team size of 4.15 authors, against 3.36 in the four peers, and 3.22 years since first recorded nineteen-journal publication, against 3.52. That history begins in 1990 and is not a complete career history. Journal-period standardisation leaves ability, seniority, team formation and topic choice uncontrolled. The lower connection rate is consistent with a network explanation, but it cannot separate the influence of networks from these other differences.

Networks can improve both the production of research and access to journals. Colussi (2018) finds that economics editors’ former doctoral students and departmental colleagues improve their publication outcomes during those editors’ terms. This concerns access within the publication process. At the *Journal of Human Resources*, authors who share a doctoral programme or former employer with the editor are less likely to be desk-rejected (Carrell, Figlio and Lusher, 2024), and authors publish more in a journal while a coauthor or colleague sits on its editorial board (Ductor and Visser, 2023). Waldinger (2010) uses the expulsion of mathematics professors in Nazi Germany to show that faculty quality affects doctoral students’ subsequent research outcomes. Azoulay, Graff Zivin and Wang (2010) finds that collaborators’ research output falls after the premature death of prominent life scientists. These studies distinguish editorial connections from the training and productive knowledge that networks transmit. They do not establish that South African manuscripts are rejected because their authors lack connections.

Doctoral study offers one route into these research relationships. Strong universities can prepare able undergraduates without themselves hosting the economics supervisors and research groups best matched to every student’s doctoral ambitions. The relevant choice is therefore the economics department and research group, rather than the university’s overall rank alone. Among South African-educated eventual authors with a known doctorate, 47.1 per cent trained abroad, compared with 89–96 per cent in Uruguay, Chile, Argentina and Colombia. These shares condition on publication and do not compare the success rates of doctoral programmes. Foreign location alone does not measure programme quality, but training in departments with active frontier researchers gives students access to advanced methods, demanding feedback and potential collaborators. The benefit of study abroad therefore depends on the department and supervisor, and on whether the student can sustain those relationships after the doctorate.

Publication incentives. A volume-placement gap appears in South Africa and 24 of the 25 comparators in the unrestricted source records; Finland is the exception. Holding base-period sources fixed reverses the gap in South Africa and most peers. Changes in the sources represented in OpenAlex therefore prevent this comparison from isolating publication incentives. A per-article subsidy predates the base period and could affect the level of placement even when subsequent

growth is comparable. Evidence on weak and predatory outlets remains relevant (Mouton and Valentine, 2017; Kerr and de Jager, 2021). In Turkey, papers supported by per-article cash payments were cited no more than unsupported papers (Tonta and Akbulut, 2020). Assessing the formula’s effect requires variation in incentives among otherwise comparable departments or cohorts.

Data access and research support. Household microdata preceded the expansion of household research about South Africa. Linked tax records subsequently enabled documented work on firms and workers. Section 8 shows little change in the firm-topic share relative to peers across the chosen window. Access can support more research when it is accompanied by the skills and time to use the data. A student needs access to the records, the methods to analyse them and sufficient time to turn an analysis into a publication.

Applied policy research. South Africa has few economists in the public service and in independent research organisations. Academic economists therefore supply much of the applied analysis on which policy draws, through university research units, commissioned studies and long-running household surveys. This work can matter even when it is not published in leading journals: when Brazilian mayors were informed of research findings, their municipalities became more likely to adopt the policy the research supported (Hjort et al., 2021). Time spent on commissioned or sector-specific projects is time not spent on the longer, riskier projects that top journals reward, although leading journals do publish South African policy research, as the household studies in Section 5 show. The publication counts in this paper measure one output of the profession; they do not show that the current allocation between the two kinds of work is wrong.

Hiring and critical mass. Departments can also add research capacity by appointing established researchers from abroad, including South Africans returning from foreign careers. Work-permit procedures and the priorities that universities attach to employment equity shape such appointments, and the salaries offered compete with those of foreign universities. Research careers are also more attractive where several active researchers work together, so a few appointments in one department can change its prospects quickly. These are hypotheses: the data here do not observe applications, offers or appointments.

Race. The exclusions described in Section 2 remain visible among leading-journal authors. I coded the population group of the 223 authors with a South African first affiliation in the nineteen journals, and of the South African-educated authors who published only from abroad, from public biographical information; the person-level coding is kept confidential. Of the domestic authors, 41 are foreign nationals. Among the 182 South Africans, 70 per cent are white, 16 per cent black African, 8 per cent coloured and 5 per cent Indian. The relevant comparison is with the people who study and publish economics in South Africa. In 2022, 68 per cent of economics students at the traditional universities were black African and 20 per cent white, and in 2020, 43 per cent of all authors of subsidy-earning publications from South African economics departments were black

African and 39 per cent white (Branson and Whitelaw, 2024). Among the 97 domestic authors in the nineteen journals in 2020–25, including foreign nationals, the corresponding shares are 16 and 58 per cent. The black African share of South African article-equivalents in these journals was zero in the 1990s, 11 per cent in the 2000s, 19 per cent in the 2010s and 18 per cent in 2020–25. It has risen, but far less than among students or in the country’s wider accredited economics output. No black South African economist, at home or abroad, has a regular top-five article in 1990–2025, and fewer than five have published in the ten-journal set. Almost all of the 30 South Africans in the group abroad are white, and none is black African. Women’s representation differs: coded from names and checked by hand, women are 34 per cent of domestic leading-journal authors and 41 per cent in 2020–25, close to their share of about a third of all authors of accredited economics publications (Branson and Whitelaw, 2024), and they hold 6 of the 17 domestic author places in the ten-journal set. These counts describe who publishes in these journals, not the ability of any group. The routes into leading-journal research, including doctoral study abroad, are themselves unequal (Section 9). Widening them is a matter of equity and would also enlarge the pool from which future researchers are drawn.

Non-academic employment. Relative returns can influence who continues in research (Murphy, Shleifer and Vishny, 1991; Philippon and Reshef, 2012; Célérier and Vallée, 2019). The large finance share of domestic publication describes those who publish; it does not show that talented students left for finance. The career mechanism can be investigated more directly by comparing marks and doctoral progression among graduates, or by relating progression to hiring conditions at graduation in the manner of Oyer (2008). Linking education, employment and publication would distinguish entry into research from retention after the doctorate.

Editorial selection and research demand. Placement also depends on the journals’ editorial choices within a hierarchical international discipline (Chelwa, 2021). A small domestic top-five presence can reflect submission patterns, editorial selection or both. Submission and decision data would separate them; publication counts cannot. Similarly, the policy-speech comparison describes stated topics rather than the institutions that commission research (Hirschman and Berman, 2014; Padayachee and Sherbut, 2011).

11 Research policy implications

The evidence motivates a practical proposal: economics departments could identify and prepare more of their strongest students for doctoral study at leading international research departments. Advanced mathematics and econometrics, research assistantships, application mentoring and financial support can make such placements attainable for a wider pool. Scholarships should target strong supervisors and research environments, rather than treating every foreign doctorate as equivalent. Such support would respond to the concerns about doctoral capacity raised in the national review (Academy of Science of South Africa, 2010).

Training abroad should be paired with opportunities to build a domestic research career: competitive return fellowships, protected research time, joint appointments and continuing collaboration with supervisors and peers. Researchers who remain abroad can contribute through co-supervision and shared projects, as diaspora researchers have done for Chinese science (Xie and Freeman, 2020). Evidence on return schemes elsewhere is encouraging but not directly transferable. Fulbright fellows required to return home were cited more often at home than comparable scientists who stayed in the United States, and continued to cite research produced there (Kahn and MacGarvie, 2016); and researchers recruited to China through a return programme later published more than comparable peers who stayed abroad, helped by larger teams and more funding (Shi, Liu and Wang, 2023). A researcher returning with advanced training may still struggle to complete demanding projects without collaborators and protected time. Funding these inputs together can make the training more productive and create opportunities to train the next cohort. External evidence on faculty quality and collaboration supports the mechanisms behind this proposal; the selected-author comparisons here identify a reason to investigate them in South Africa. They do not estimate the return to a scholarship or establish that foreign training is the binding constraint. Funders could monitor placements, domestic appointments, sustained collaborations and repeated five- and ten-journal publication. Estimating a programme’s effects would additionally require a credible comparison group, such as similarly qualified applicants near a funding cutoff where the award rule permits that design.

Administrative-data programmes can combine access with research support. Methods training, funded data construction, reliable documentation and predictable proposal-review times reduce different costs of using the same records. Secure remote access, where technically and institutionally feasible, would reduce travel costs. Evaluating these programmes requires verified project and publication records that distinguish the datasets used and the authors’ locations. Tracking published topics alone would miss some uses of the records.

Appointments can build on these investments. A small number of research-active appointments in one department, including returning South Africans, may do more than the same number spread across many. Administrative data on students, staff and publications, of the kind already used to track gender in the South African academic economics pipeline (Branson and Whitelaw, 2025), would allow such appointments and their effects to be monitored.

Universities can also recognise data construction, replication and research quality in promotion and funding. Departmental assessment should track sustained contributions and the development of research careers alongside publication volume and the university’s overall standing. Competitive calls on policy questions can create opportunities to apply that capacity, provided they leave researchers free to determine their conclusions.

12 Conclusion

Economists based in South Africa have a very small presence in the top journals. Only 2 regular top-five articles and 11 in the specified ten-journal set have a domestic first affiliation over 1990–2025. Chile, Brazil, Argentina and Colombia have substantially more, and holding author numbers equal preserves the gap with Chile. Counting economists educated in South Africa who work abroad does not close the gap: the identified group has six regular top-five articles, against 108 for Chilean-educated economists. Leading-journal authors in South Africa remain predominantly white, far more so than economics students or the authors of the country’s wider accredited economics output. South Africa’s full-period top-five share ranks 24th of 26 under four journal denominators, although recent-period ranks vary. Its internationally recognised universities provide a setting in which to examine the economics result, rather than a measure of expected departmental output.

Recorded publication has grown, chiefly through outlets beyond the top tier. Much of the measured volume increase comes from journals absent from the country’s base-period records, so growth ranks depend on the source frame. Foreign-based authors still hold the majority of leading-journal authorship about South Africa. That work concentrates on households, while domestic research concentrates on finance. The relatively small measured firm portfolio suggests scope for further work, but classification uncertainty precludes a precise estimate of its shortfall against peers.

Among eventual leading-journal authors with known doctoral histories, the South African-educated group has a smaller foreign-doctoral share than Uruguay, Chile, Argentina and Colombia. Domestic authors also have fewer recorded prior collaborations with established top-five authors than those in Argentina, Brazil, Chile and Colombia, including after matching their journal-period mix. These selected-author comparisons do not identify the effects of training or networks. Together with external evidence on faculty quality and collaboration, they motivate preparing more strong students for leading international economics doctoral departments and evaluating support for their subsequent research careers. Such support can combine time to develop projects, continuing collaboration and opportunities to train younger researchers. It need not come at the expense of the applied policy research that South African economists already supply, but it does require protecting time for longer projects. The objective is for successive cohorts of economists to contribute more to the discipline and to understanding – and, ultimately, addressing – the country’s economic problems.

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A Additional measurement details

Publication denominators. Table 5 uses regular top-five fractional article-equivalents in every numerator. Its denominators contain nineteen journals, eighteen without JAE, seventeen without JAE and *World Development*, and the specified ten. All exclude historical AER proceedings. Countries retain the main table’s order; ranks are descending shares, with ties receiving the same minimum rank. The full-period South African rank is 24 under every denominator. For 2020–24 the corresponding ranks are 22, 22, 16 and 19. These are placement shares within selected journal sets, not publication probabilities per economist or measures of research value.

National sample flow. Table 6 starts from the same 15,214 unique work IDs as the output series. Among the 13,862 journal-confirmed works, 11,416 have abstracts and were submitted for classification; three failed classification and one was removed after validation, leaving 11,412. Zero domestic-first- affiliation weight removes 915 further records from positive-weight topic totals. The record-level ledger archives each inclusion flag. The 21 non-journal-primary DOI matches are already excluded by journal scope. The final column divides classified domestic article weight by all journal-confirmed domestic weight; the output-growth calculations do not condition on classification or abstract availability.

Topic benchmark specification. For topic k , comparator country c and period t , the outcome is the country-period mean topic probability, weighted within the cell by sampling expansion and domestic first-affiliation fractions. The conditional mean is

$$E[s_{kct} \mid x_{ct}] = \Lambda(\alpha_k + \beta_k \log y_{ct} + \gamma_k \log P_{ct} + \delta_{kt}),$$

Table 5: Top-five shares under alternative journal denominators.

Country	19 journals	18 journals	17 journals	17 rank	10 journals	10 rank
Russia	35.7	35.7	37.7	1	50.7	5
Israel	28.9	28.9	30.4	3	50.7	4
Czechia	20.3	20.8	23.5	5	35.6	15
Turkey	19.1	19.3	26.8	4	73.2	1
South Korea	16.8	17.0	21.3	6	52.4	2
Portugal	15.0	15.2	16.8	8	34.5	16
Kazakhstan	15.0	15.0	32.1	2	42.9	11
Greece	14.7	15.4	16.6	10	45.6	9
Poland	11.2	12.0	14.2	14	38.0	12
Argentina	11.1	11.2	14.8	13	47.4	7
India	10.1	10.1	17.6	7	46.3	8
United Arab Emirates	10.0	10.0	14.9	12	37.5	13
Brazil	9.6	9.6	15.7	11	43.5	10
Chile	8.4	8.7	12.0	18	29.9	17
Mexico	7.5	7.6	13.5	15	36.2	14
Uruguay	7.3	7.3	12.5	16	27.4	19
Finland	6.5	6.9	9.2	22	21.7	22
New Zealand	6.3	6.8	10.0	19	21.9	21
Thailand	3.3	3.3	16.7	9	47.6	6
Bangladesh	2.7	2.7	9.8	20	16.1	23
Colombia	2.6	2.7	4.8	23	25.8	20
Indonesia	2.3	2.3	12.4	17	52.2	3
Vietnam	1.5	1.5	9.5	21	28.6	18
South Africa	0.5	0.7	3.8	24	11.7	24
Egypt	0.0	0.0	0.0	25	0.0	25
Malaysia	0.0	0.0	0.0	25	0.0	25

Table 6: National sample construction by period.

Period	Collected	Journals	With abstracts	Classified journals	Positive SA weight	Weight retained (%)
1990-1994	406	337	207	207	202	61.4
1995-1999	435	395	257	257	242	64.1
2000-2004	777	635	431	431	405	67.3
2005-2009	1121	936	742	742	699	80.2
2010-2014	2451	2106	1747	1747	1650	85.1
2015-2019	3426	3134	2491	2490	2298	84.0
2020-2024	5313	5084	4392	4389	3950	89.0
2025-2025	1285	1235	1149	1149	1051	95.8
All years	15214	13862	11416	11412	10497	84.4

where y_{ct} and P_{ct} are arithmetic means of annual real PPP income per head and population within the period, logged after averaging. The twelve regressions use binomial fractional-logit quasi-likelihood and give each country-period cell equal weight. Periods run from 1990–94 to 2020–24, followed by 2025 alone. The 25 comparators supply 200 cells; the UAE’s 2025 income observation is missing, leaving 199 estimation cells. South Africa never enters the estimation sample.

For target portfolio r , predicted shares are aggregated as

$$\hat{s}_{kr} = \sum_t \omega_{rt} \Lambda \left(\hat{\alpha}_k + \hat{\beta}_k \log y_{ZA,t} + \hat{\gamma}_k \log P_{ZA,t} + \hat{\delta}_{kt} \right), \quad \omega_{rt} = \frac{\sum_{a \in (r,t)} w_a}{\sum_{a \in r} w_a}.$$

Domestic weights are fractional first affiliations; about-South-Africa articles have unit weight. The two portfolios therefore have different time weights and predictions. Ratios divide their weighted observed shares by these predictions. Separate marginal logits do not impose adding up across topics: baseline predicted shares sum to 1.0076 for domestic output and 1.0005 for work about South Africa. They are topic-specific benchmarks, not a jointly estimated allocation of an entire portfolio.

Table 7 reproduces the original predictions and then removes the income and population co-variates, replaces period indicators with a linear time trend, or uses period means from eight matched countries. Matching uses Euclidean distance from South Africa in mean log income and population, standardised across the 26 countries, without topic outcomes. The nearest eight are Colombia, Thailand, Egypt, Turkey, Mexico, Vietnam, Brazil and Argentina. They receive equal weight within period, with the same target time weights as the baseline. The final column omits one comparator at a time from the original specification. Entries are model ratios before classifier correction. These checks assess benchmark choice; they do not remove selection through missing abstracts.

Table 7: Topic ratios under alternative benchmarks.

Portfolio	Topic	Baseline	Period only	Linear time	Matched eight	Leave-one-out
Domestic	Firms	0.60	0.64	0.61	0.54	0.56–0.64
Domestic	Households	1.06	1.29	1.09	1.41	1.04–1.12
Domestic	Finance	1.37	1.25	1.38	1.46	1.30–1.53
About SA	Firms	0.61	0.64	0.61	0.57	0.58–0.65
About SA	Households	4.02	4.70	4.13	5.00	3.92–4.22
About SA	Finance	0.24	0.22	0.26	0.27	0.23–0.27

Table 8 reports the primary corrected level estimates and their 95 per cent stratified-bootstrap intervals. The difference estimator is not constrained to the probability simplex, so an interval may extend below zero; this indicates imprecision rather than negative topic mass. The classification intervals condition on the supplied human labels and do not cover missing abstracts or cross-genre transfer to speeches.

Table 8: Corrected topic shares and intervals.

Topic	Domestic (%)	About South Africa (%)
Firms	4.4 [-0.2, 9.3]	4.1 [2.1, 6.2]
Labour	3.0 [-0.8, 7.1]	9.5 [5.1, 13.9]
Households	2.4 [-2.2, 7.6]	22.8 [16.8, 28.5]
Fiscal	6.0 [2.8, 9.5]	9.4 [6.4, 12.7]
Education	3.3 [1.7, 5.7]	6.8 [5.7, 7.6]
Health	11.7 [6.2, 17.5]	10.8 [7.7, 14.5]
Macroeconomics	14.9 [8.4, 21.5]	9.0 [6.9, 11.1]
Trade	4.9 [1.3, 8.6]	6.1 [4.1, 8.2]
Finance	27.4 [20.8, 33.3]	3.4 [2.3, 4.8]
Institutions	10.1 [4.7, 15.9]	8.3 [6.2, 10.2]
BEE	0.3 [-0.2, 0.6]	4.4 [2.9, 6.2]
Other	11.7 [7.5, 17.0]	5.4 [2.6, 8.6]

Source representation across countries. Table 9 extends the Indonesia calculation to all 26 countries. For each country, the base source set consists of source IDs represented among journal-confirmed records in 2000–09. Base-source growth divides 2015–24 article weight in those sources by total base-period weight. The remaining later weight gives the newly represented-source percentage. Placement F retains its fixed nineteen-journal definition; both ranks order $\log(F \text{ growth}) - \log(Q \text{ growth})$ from high to low. South Africa is fully enumerated; sampled comparator cells use their frozen expansion factors. A source absent from a sampled base cell may have existed and published country-affiliated work then, so this restriction is a sensitivity on observed source support rather than a complete balanced journal census. It does not date journal foundation or database entry.

Unresolved comparator affiliations. Table 10 reports the unweighted proportion of unresolved author positions in journal-confirmed comparator records for each comparison window. The growth range assigns all unresolved fractional mass to the domestic country in the early window and none in the late window, then reverses those assignments. Resolved affiliations are held fixed. The last column instead assigns an author domestically when any recorded affiliation is domestic. These calculations preserve the article and sampling weights. Unresolved rates vary across countries and periods; neither design weights nor the South African audit establish cross-country reconstruction accuracy. The ranges address unknown positions, not errors in apparently resolved ones, and are separate from the source-support sensitivity.

Journal-conditioned network comparisons. Table 12 reports prior connected positions over all identified positions, with percentages in parentheses, for three baskets. The prior history remains the full nineteen-journal bank in each case. For the journal-period standardisation, the four Latin American countries are pooled; their connection rate in each journal and five-year period is weighted

Table 9: Publication growth within base-period sources.

Country	All Q	Base-source Q	New sources (%)	F	Original rank	Base-source rank
Finland	2.09	0.82	60.6	3.21	1	4
United Arab Emirates	8.43	1.36	83.8	6.23	2	3
Uruguay	6.59	0.35	94.7	4.87	3	1
Czechia	5.35	1.22	77.2	3.14	4	8
Chile	4.28	1.39	67.5	2.30	5	12
South Africa	4.48	1.27	71.6	2.37	6	10
Israel	1.49	0.64	57.1	0.74	7	18
New Zealand	1.68	0.93	44.3	0.77	8	21
Brazil	4.99	1.61	67.7	2.20	9	15
Colombia	10.48	1.73	83.5	4.55	10	6
South Korea	4.00	1.77	55.8	1.72	11	19
Poland	5.23	1.83	65.1	2.18	12	17
Portugal	3.73	1.09	70.9	1.50	13	14
Argentina	5.01	0.88	82.5	1.37	14	13
Egypt	13.60	1.35	90.1	3.49	15	7
Thailand	8.12	0.89	89.0	1.89	16	9
Mexico	6.34	1.38	78.1	1.23	17	20
India	10.51	2.32	77.9	1.78	18	23
Vietnam	42.24	1.87	95.6	7.08	19	5
Malaysia	11.76	2.02	82.8	1.64	20	22
Russia	14.58	1.32	90.9	1.70	21	16
Bangladesh	17.00	1.06	93.8	1.83	22	11
Greece	2.78	1.12	59.8	0.30	23	26
Turkey	15.07	1.96	87.0	1.45	24	24
Kazakhstan	28.10	0.19	99.3	1.80	25	2
Indonesia	70.94	2.36	96.7	1.56	26	25

Table 10: Unresolved affiliations and output-growth sensitivity.

Country	Unknown, early (%)	Unknown, late (%)	Q growth range	Any affiliation
Argentina	5.3	7.3	4.69–5.43	4.98
Bangladesh	7.1	3.7	15.58–17.75	14.19
Brazil	6.0	5.7	4.74–5.27	4.76
Chile	6.4	5.9	3.95–4.64	4.40
Colombia	8.2	7.4	9.66–11.27	10.40
Czechia	5.7	4.4	5.07–5.65	5.01
Egypt	5.0	4.0	12.84–14.21	12.90
Finland	3.1	3.4	2.03–2.15	2.22
Greece	2.9	2.9	2.72–2.84	2.79
India	5.3	3.7	10.14–10.90	9.97
Indonesia	6.0	6.1	68.37–74.53	68.85
Israel	2.5	3.2	1.45–1.53	1.48
Kazakhstan	8.9	10.5	26.18–31.45	25.36
Malaysia	3.4	4.8	11.31–12.48	11.79
Mexico	9.2	7.8	5.77–6.84	6.11
New Zealand	4.5	2.1	1.62–1.71	1.71
Poland	5.0	2.6	5.07–5.33	5.28
Portugal	3.5	3.4	3.62–3.91	3.88
Russia	6.1	7.0	14.25–15.39	14.04
South Korea	4.0	2.3	3.87–4.11	3.98
Thailand	3.5	2.0	7.79–8.33	7.87
Turkey	2.4	3.2	14.75–15.55	15.11
United Arab Emirates	1.7	2.4	8.25–8.71	8.50
Uruguay	9.1	7.4	5.59–7.35	6.33
Vietnam	15.8	5.3	37.85–44.30	36.09

Table 11: Doctoral routes and missing-country bounds.

Country	Identified	Known PhD	Unknown	Foreign	Foreign, known (%)	Full-frame range (%)
Uruguay	38	36	2	32	88.9	84.2–89.5
Chile	202	195	7	181	92.8	89.6–93.1
Czechia	65	61	4	18	29.5	27.7–33.8
Argentina	215	208	7	192	92.3	89.3–92.6
Colombia	152	136	16	130	95.6	85.5–96.1
Turkey	166	163	3	157	96.3	94.6–96.4
South Africa	111	102	9	48	47.1	43.2–51.4
Poland	64	63	1	35	55.6	54.7–56.2
Brazil	281	274	7	180	65.7	64.1–66.5
Mexico	84	77	7	72	93.5	85.7–94.0
India	518	498	20	414	83.1	79.9–83.8
Malaysia	8	8	0	6	75.0	75.0–75.0
Thailand	12	11	1	10	90.9	83.3–91.7
Egypt	16	16	0	15	93.8	93.8–93.8
Indonesia	29	29	0	26	89.7	89.7–89.7

by South Africa’s share of positions in that cell. Every South African cell has peer observations. The estimate holds journal and period composition constant, without conditioning on team size, seniority or ability. It remains a comparison among authors selected into publication.

Table 12: Prior connections by journal basket, 2015–24.

Country	19 journals	17 journals	10 journals
Argentina	31/90 (34.4)	27/59 (45.8)	12/22 (54.5)
Brazil	55/245 (22.4)	52/153 (34.0)	33/73 (45.2)
Chile	63/218 (28.9)	58/176 (33.0)	29/82 (35.4)
Colombia	29/203 (14.3)	22/92 (23.9)	8/27 (29.6)
Czechia	29/82 (35.4)	29/71 (40.8)	23/46 (50.0)
South Africa	12/171 (7.0)	2/32 (6.2)	0/14 (0.0)

University standing and recent publication. Table 13 retains Table 2’s country order and reports THE 2026 institution counts at three thresholds, alongside economics publication in 2020–24. The publication entries are distinct articles with a domestic first affiliation; the percentage uses fractional article-equivalents in both numerator and denominator. All three journal sets exclude historical AER proceedings. The period calculations, rank bands and institution names are retained in the replication archive. South Africa’s recent share is 1.9 per cent, ranked 22nd. Excluding the two development journals changes the recent rank to 16th, compared with an unchanged full-period rank.

THE’s overall ranking combines teaching (29.5 per cent), research environment (29), research quality (30), international outlook (7.5) and industry (4). The research-quality component uses publications from 2020–24 and citations through 2025 (Times Higher Education, 2026). These institutional measures are broader than elite-journal economics publication. Matching the publication window improves the timing of the comparison. The ranking remains a measure of contemporary institutional standing, not a control for university quality throughout 1990–2025.

Institutions are assigned to the country or territory recorded by THE and count once per listed institution. Exact ranks and ties are included through the stated threshold; banded ranks are included when the band’s upper limit is at or below it. Thus every institution in the 401–500 band is counted in the top 500, including ties at its boundary. Unranked reporters are excluded. THE’s data submissions, publication requirements and subject-coverage rules affect representation. A zero is verified against the full country listing and means no institution represented within that cutoff. Unlisted universities are outside the measure. Uruguay has reporter entries but no ranked institution in this edition.

The institution ledger uses a pinned machine-readable transcription of THE’s published results, checked against independently hosted global and Russian rank tables and official THE profiles for the principal institutional comparisons. Source locations, access dates, verification status and file hashes are archived. Country listings were found for all 26 countries. At the top-200, top-500 and top-1000 cutoffs, respectively, 18, 16 and 13 peers have fewer represented institutions than South Africa but more regular top-five articles over 1990–2025. The corresponding counts for 2020–24 are eleven, nine and six. These counts do not assume a linear relationship between university numbers and top-five articles.

Topic classifier. Every corpus is classified by the same fixed twelve-topic instrument (Claude Haiku 4.5, Batch API, temperature zero), which returns a probability vector over the twelve classes summing to one; the prompt is archived with the replication package. Excluded records are listed in the replication files. Of 12,587 South African-affiliated articles submitted, 12,584 returned valid classifications; of 126,904 comparator articles, 126,876; all 246 about-South-Africa articles and the 7,496 retained policy paragraphs classified cleanly. Independent validation identified one mismatched South African record and one non-academic comparator record, so the broad portfolios contain 12,583 and 126,875 records. The journal-and-ISSN rule retains 11,412 South African records and 108,284 comparator country-records before embedding availability; of the South African records, 10,497 have positive first-affiliation weight.

Topic maps and intervals. Figure 4 projects SPECTER2 article embeddings onto a fixed two-dimensional coordinate system trained on comparator articles. SPECTER2 base embeddings with the SPECTER2 adapter use a 512-token input limit. Randomised PCA reduces them to 50 dimensions, followed by two-dimensional UMAP with 30 neighbours, minimum distance 0.10 and cosine distance. Both stages use seed 73; UMAP also uses transform seed 73 and one thread. South African articles are transformed into that fixed comparator space. Coordinates are inherited from

Table 13: University rankings and recent economics publication.

Country	THE top 200	THE top 500	THE top 1000	Regular top five	Ten journals	Top-five share (%)
Russia	1	3	15	23	44	45.1
Israel	0	3	7	21	58	18.3
Czechia	0	1	3	4	17	7.5
Turkey	0	4	15	10	13	26.4
South Korea	6	14	26	11	18	17.5
Portugal	0	3	9	7	21	15.9
Kazakhstan	0	1	1	2	3	31.6
Greece	0	1	6	0	3	0.0
Poland	0	0	4	1	2	3.6
Argentina	0	0	0	2	6	4.9
India	0	4	43	19	40	13.6
United Arab Emirates	0	7	8	3	8	9.4
Brazil	0	2	8	11	30	8.4
Chile	0	1	1	8	32	6.0
Mexico	0	0	2	4	22	4.3
Uruguay	0	0	0	0	4	0.0
Finland	2	9	9	8	25	7.5
New Zealand	1	4	8	2	7	4.1
Thailand	0	0	3	1	1	7.4
Bangladesh	0	0	5	3	12	4.8
Colombia	0	0	0	4	15	2.8
Indonesia	0	0	1	2	3	7.0
Vietnam	0	0	4	1	2	2.9
South Africa	1	4	8	2	7	1.9
Egypt	0	0	9	0	2	0.0
Malaysia	0	7	13	0	2	0.0

the broader abstract-covered frame; journal restriction reweights its hexagons without refitting the projection. For hexagon h , let Z_h and W_h be domestic and comparator article mass, with totals Z and W . The colour uses

$$\tilde{L}_h = \frac{Z_h + 10}{(W_h/W)Z + 10}.$$

The grid size is 42. Displayed $\log_2 \tilde{L}_h$ is clipped to $[-2, 2]$; cells with fewer than thirty comparator records are neutral. Both portfolios use fractional first-affiliation weights and comparators also carry sampling expansion weights. In Figure 3, the dark interval is the envelope of country-block and author-block bootstraps, each with 999 replicates. The pale classifier interval uses 999 resamples of validation labels. Axis annotations give domestic topic-probability mass (Nw) and the number of domestic articles with positive topic probability (N); BEE is race, ownership and affirmative action. Figure 9 separates the original ratios and bootstrap intervals from the symmetric correction and its classifier-error bands. Bands reaching the left edge use the share floor described below.

Journal scope. The collection query used OpenAlex’s `type:article`, which also includes records outside journals. The main national sample requires a primary source of type `journal` and a recorded ISSN, applied identically to comparators. It retains 13,862 of 15,214 national records. Restricting the classified sample changes South African topic shares by at most 0.6 percentage points and comparator shares by at most 0.2.

Journal-tier and network audit. The five-, ten- and nineteen-journal comparisons use nested sets and remove historical AER proceedings throughout. The replication files retain the country–journal counts and each South African ten-journal author position. Publisher affiliation order corrects Gary Magee’s EJ article (DOI 10.1111/eco.12497) to Australia and Michael Carter’s EJ article (DOI 10.1093/ej/ueae012) to the United States. Both works are absent from the national economics-field pull and from the about-South- Africa sample; their corrections affect placement, not the national output or topic weights. The corpus uses indexed publication years, which can precede the issue year for online-first articles.

For Table 14, a connection is available only after both the first observed coauthorship and the partner’s first regular top-five publication. The global nineteen-journal bank supplies the histories from 1990; author IDs are required. The comparison includes domestic positions in 2015–24 and excludes historical AER proceedings. In 2015–24, IDs are missing for one located slot each in Brazil, Chile and Colombia, and none in South Africa, Argentina or Czechia. The statistic weights publication positions equally; a separate replication table averages each author’s position-level connection share and then gives authors equal weight. Neither measure treats a new collaboration on the focal publication as a pre-existing connection.

Indonesia source audit. The base-period cells are complete frozen pulls; the 2015–19 and 2020–24 cells each sample 1,500 works from recorded populations of 12,890 and 40,307. Their expansion factors are 8.59 and 26.87. The audit reproduces these weights, checks unique work IDs and non-

Table 14: Prior collaboration with top-five authors, 2015–24.

Country	Author positions	Prior connection	Share (%)
South Africa	171	12	7.0
Chile	218	63	28.9
Colombia	203	29	14.3
Brazil	245	55	22.4
Argentina	90	31	34.4
Czechia	82	29	35.4

missing DOIs, and reconstructs the plotted totals. Among 1,679 source IDs represented in later journal-confirmed output, 1,627 are absent from the base records and account for 96.7 per cent of later weight. Base-source output grows 2.36 times. Sources absent from a country’s base records need not be newly founded or newly indexed; this is a fixed-source sensitivity, not a dating of journal entry into OpenAlex. The 70.94 growth ratio compares pooled output in 2015–24 with 2000–09.

The metadata rule can admit proceedings series: *KnE Social Sciences*, which its publisher describes as publishing conference proceedings (Knowledge E, 2026), contributes 199 later article-equivalents. Removing it reduces Indonesia’s growth to 70.62 times, so this identified scope issue does not explain the outlier. Most of the increase comes from sources absent in the base period. Excluding Indonesia altogether leaves South Africa’s growth rank at six and its residual from the comparator line at approximately 0.46 log points. The replication archive includes sampled titles, journal names, source-level weights and the finite-population sampling interval.

Research on South Africa and abstract coverage. Of 430 prefilter candidates, 246 were classified as about South Africa (43 as multi-country studies); two corrupt OpenAlex records are excluded by documented overrides. Table 16 reports abstract coverage by journal and period after the Semantic Scholar recovery. The prefilter uses titles, abstracts and author affiliations. An article without an abstract is missed only if neither its title nor its author affiliations would trigger it; among retained articles with abstracts, 22 per cent were detectable only through the abstract. The corrected series in Figure 1 adds, within each journal and period (1990–99, 2000–09, 2010–19 and 2020–25), the number of uncovered articles times the about-South-Africa rate among covered articles times that share. A higher-missingness scenario that treats every uncovered article as detectable only through its abstract reaches 18.9 per thousand in 2000–04; its 2005–09 value is 8.7. This high-missingness scenario is distinct from the main journal-by- decade correction, which peaks at 8.2 in 2005–09. It is a scenario under an assumed missing-article rate, not a formal upper bound on all articles about South Africa. A series restricted to the six journals with coverage above 90 per cent in every period peaks at 2.9 per thousand in 2005–09 and is between 1.4 and 1.7 afterwards. Table 17 and Figure 8 give the journal-subset series. Table entries are articles about South Africa

per 1,000 corpus articles. The fixed comparison uses 1990–2008 and 2009–2025; the post-peak comparison uses 2005–09 and 2010–25. Ratios are late over early. Exact conditional Poisson intervals are reported for the fixed windows; selected-peak contrasts are descriptive and have no selection-adjusted intervals. The “top five and general” subset uses the same ten journals as Table 2, but retains historical AER proceedings. The balanced subset excludes JEEA, AEJ Applied and AEJ Macro, which enter after 1990. Figure 8 shows five-year rates with Wilson intervals and article counts, alongside the full-set coverage scenarios and restrictions. The latter include a correction using journal-by-five-year cells in addition to the main correction using journal-by-decade cells.

Reconstruction of comparator first affiliations. For comparator records, I take the author’s first-listed institution from the ordered affiliation field in the OpenAlex deposit. The code identifies its country from the record’s institution list. If that fails, it uses the authorship’s single country tag, then the trailing country name in the raw string. Of 438,003 author positions, 6.3 per cent remain unresolved and carry zero weight.

The nineteen-journal bank stores country tags as an unordered set, so it requires a separate reconstruction from raw affiliation strings. A single tagged country is used unless the raw string names a different country. For multiple-country records, the code takes the earliest mention in the first raw string of either a tagged country or an institution with a known country. The institution lookup uses single-country author positions and comparator institutions. Strings with deposit-order labels are first reordered by those labels.

I checked the rule against 816 adjudicated South African author positions. Before the additional publisher corrections reported above, agreement was 94.5 per cent overall and at least 90 per cent in every bin with thirty or more slots; false positives and false negatives are equal within three; and reconstructed South African frontier growth was 1.09 times the adjudicated figure in that validation. The affiliation-validation plan pre-specified a 95 per cent threshold; the first run reached 94.2 per cent and inspection showed the residual disagreements arise where OpenAlex stores raw strings in a different order from the Crossref deposit. The threshold was relaxed to 94 per cent and the change is recorded in the script. After the publisher corrections, the reconstructed South African author count is 223 and the adjudicated count is 221; the corresponding identified-author article-equivalents round to 118 and 119.

Cross-country publication growth. Table 18 lists the country-level growth rates behind Figure 6. In the table, Q is journal-confirmed output, F nineteen-journal placement and T top-decile citation impact. Growth compares 2015–24 with 2000–09, except that T ends in 2011–20. The table reports $g = \log(F \text{ growth}) - \log(Q \text{ growth})$ and its descending rank. Base F is fractional placement in 2000–09; a dagger marks a base below five article-equivalents, excluded from the fitted line. Comparator output is weighted by bin-level expansion factors multiplied by domestic first-affiliation fractions; 133 of 200 cells are full enumerations. South Africa’s output uses the Crossref-corrected fraction; the uncorrected OpenAlex fraction gives the same growth to one decimal. The adjudicated South African placement growth is 2.2 against 2.4 from the reconstructed

rule. The fitted line has slope 0.20 and R^2 of 0.09 on 18 countries; the prediction interval for a single country is -1.31 to +1.31.

With the minimum base set at three (22 countries) South Africa’s residual is +0.40 and its rank 5; at ten (16 countries) the residual is +0.52 and the rank 3; a fit weighted by base placement on all comparators gives +0.53. Removing unresolved author slots from the placement denominators gives $g = -0.66$ and rank 6; using the adjudicated South African placement series gives $g = -0.73$ and rank 7 of 26 (5 of 19 with a usable base).

The publication bootstrap uses the union of South African national and frontier publication IDs in each comparison window. It resamples IDs jointly, retaining zero frontier contributions, and recomputes both growth rates and their difference. It conditions on the frozen bibliographic coverage and affiliation classifications. The size-matched concentration draws in Section 7 sample 221 authors without replacement from Chile’s reconstructed author pool 999 times and record the top-ten share, the count of authors with three or more articles, the article-equivalents and the top-five author positions of each draw.

Policy speeches. From the government record, 67 genuine State of the Nation and Budget addresses survive an explicit title-and-URL inclusion rule; eleven derivative records are excluded and listed in the archived QA file. Two retained addresses (the 2007 and February 2009 States of the Nation) have no classified paragraphs, which leaves the 65 analytical addresses; the 1994 and 1995 Budget speeches are absent from the source, and no genuine 2015 State of the Nation address survives the rule. Low-confidence paragraph classifications, 6.9 per cent of the corpus, are retained.

Classifier validation and correction. A random sample of 400 classified articles with recorded inclusion probabilities (120 South African-produced, 120 about South Africa, 160 comparator) was hand-coded independently and blind to the model output. Three records flagged by the coder and one previously registered corrupt record are excluded, leaving 396 scored records. Probability-weighted strict agreement is 61.2 per cent; Table 15 gives design-weighted precision and recall by topic from the 396 scored records. Secondary-accepted precision also counts agreement with the coder’s secondary topic label.

The prediction-powered correction applies the difference estimator of Ludwig, Mullainathan and Rambachan (2025) within sampling strata. On the journal-confirmed main sample, it gives domestic finance 27.4 per cent, household 2.4 and firms 4.4. A stratified bootstrap with 1,999 replicates gives a household interval from -2.2 to 7.6 per cent and a firm interval from -0.2 to 9.3 per cent. The corrected domestic household share cannot be distinguished from zero. The prediction-powered estimate is the primary corrected level estimate, conditional on the probability sample and the independent labels.

The symmetric correction estimates the mean-probability confusion matrix from the scored rows. It inverts the matrix under non-negativity and adding-up constraints for South Africa, the about-South-Africa corpus and every comparator country-bin, then refits the twelve fractional logits and recomputes the ratios. The classifier bands in Figures 3 and 9 use 999 resamples of validation

rows within strata. They assume that the estimated probability matrix transfers across countries and periods. For the 244-article about-South-Africa corpus the inversion reaches the zero boundary for firms, macroeconomics and the residual class in most replicates, so those bands are truncated at a 0.1 per cent share floor and should be read as poorly determined. Topics with fewer than twenty human-coded rows are race, ownership and affirmative action (10) and labour (16).

Researcher counts and doctoral training. Figure 7 uses machine-identified national counts. Table 11 reports the known, unknown and foreign doctoral counts and full-frame route bounds. These bounds allocate unknown PhD countries within the identified frame; they do not bound the omitted population of economists. Cohort counts condition on home undergraduate education, a known PhD completion year and eventual publication in the journal set.

The companion census’s generalised-difference correction adds stratum-level mean validation residuals to machine counts. Let Y denote true joint inclusion, M machine inclusion, H the six machine strata, R whether the audit outcome is resolved and C country. Applying resolved residuals to unresolved records requires

$$E[Y - M \mid H, C, R = 1] = E[Y - M \mid H, C, R = 0].$$

Pooling residuals across countries additionally requires their conditional means to transfer between countries within H . Audit design weights address selection into the audit, not selection into resolved outcomes. With 316 of 608 home-education outcomes unresolved, neither equality is established by the evidence. The South African corrected rate of 2.93 per million and its 2.58–3.28 validation-sampling interval are consequently conditional calculations, retained in the replication archive rather than used for the main comparison. Adjudication error and non-random missingness are outside those intervals. In held-out-country diagnostics, only 16 of 96 test cells have at least five resolved records; these checks concern resolved cases and do not validate transfer to unknown biographies.

Publication types. Table 19 lists the South African author positions in the top-five journals, with publication types checked against publisher records. Historical AER proceedings are identified from the May issue: issue 2 through 2010, issue 3 in 2011–13 and issue 5 in 2014–17. Sixteen retained AER records lack issue metadata and remain in the sensitivity frame. The inherited corpus also contains 22 records titled “The American Economic Review” rather than individual research titles. Removing these identified front-matter records from the attention denominator is recorded as a sensitivity in the replication tables. The common frame is an indexed journal-publication measure, with publication types distinguished in the top-tier interpretation. Table 20 repeats the fixed-window and post-peak attention comparisons after removing proceedings from both counts and article exposures. Only fixed-window rows report conditional Poisson intervals; the peak-based comparison is descriptive.

Administrative-data comparison. The country-year panel covers 2008–2024. Within countries, firm-topic probabilities are weighted by fractional first affiliations and comparator sampling expansion factors; country-year cells enter with equal weight. Table 21 reports the 2016-cutoff difference in differences, an event-study endpoint relative to 2015, a synthetic-control comparison, and a false 2012 cutoff within the earlier window. Country-placebo tail fractions rank absolute estimates across 26 assignments. They are descriptive because access did not occur through random assignment and the donor countries are not verified untreated units. The reported rank counts assignments whose absolute estimate is at least as large as South Africa’s, and the tail fraction divides that rank by 26. Estimates are in percentage points. Figure 11(a) plots the domestic firm share and equal-country comparator mean. Panel (b) shows South Africa’s relative annual share and the grey placebo-country series, each centred on its own 2008–15 mean. The dotted line marks the 2016 comparison cutoff. The event-study table instead uses 2015 as its reference year. The synthetic fit uses pre-2016 firm shares and mean household and finance shares. The alternative-cutoff results and pre-period fit are recorded in the replication outputs. A non-rejection is not used to infer a minimum detectable effect or a bound obtained by dividing by classifier recall.

Table 15: Classifier precision and recall by topic.

Topic	Precision	Recall	Precision (secondary accepted)	Human positives	Model positives
Firms	0.55	0.74	0.55	24	38
Labour	0.48	0.96	0.48	16	31
Households	0.40	0.55	0.41	28	33
Fiscal	0.65	0.89	0.65	27	30
Education	0.65	0.96	0.65	26	30
Health	0.81	0.89	0.81	47	44
Macro	0.36	0.71	0.36	31	42
Trade	0.55	0.54	0.63	26	31
Finance	0.89	0.89	0.89	42	42
Institutions	0.50	0.59	0.50	34	31
BEE	0.49	0.95	0.50	10	20
Other	0.65	0.54	0.65	35	24

Table 16: Abstract coverage by journal and period.

Journal	1990–99	2000–09	2010–19	2020–25	All years	Articles
JME	14.8	7.3	31.0	50.7	23.8	2,720
JIE	9.5	9.7	34.7	49.1	25.9	2,676
JPubE	10.2	11.6	37.3	59.3	30.0	3,704
JDE	11.5	11.1	40.4	53.1	30.0	3,291
WD	7.9	8.4	41.5	65.7	33.9	7,209
ECTA	82.2	61.3	67.1	67.5	68.5	2,911
JPE	78.2	69.4	79.3	80.9	77.2	2,317
AER	60.8	84.2	83.5	98.0	81.5	6,864
JOLE	91.6	87.1	86.4	79.3	86.2	1,355
JAЕ	86.2	95.7	95.6	96.0	93.6	1,164
EJ	92.3	94.0	97.7	97.9	94.5	4,786
JEEA	NA	88.9	97.6	98.6	94.7	1,465
JHR	92.8	92.7	97.8	98.4	95.5	1,849
QJE	98.6	92.4	97.4	98.9	96.6	1,625
AEJApp	NA	84.0	96.1	99.7	96.7	725
AEJMac	NA	87.9	95.4	99.6	96.8	600
REStat	96.8	97.8	99.0	99.3	98.2	3,025
REStud	96.8	99.1	99.8	97.5	98.4	1,950
RAND	97.7	98.4	99.7	99.5	98.7	1,285

Table 17: Research on South Africa by journal subset and period.

Subset	Window	Early	Late	Rate ratio	95% interval
All nineteen journals	Fixed window	4.22	5.16	1.22	[0.94, 1.60]
All nineteen journals	After peak	7.11	5.01	0.71	—
Top five	Fixed window	2.14	1.27	0.59	[0.25, 1.38]
Top five	After peak	3.63	0.97	0.27	—
Top five and general	Fixed window	1.50	1.82	1.22	[0.64, 2.36]
Top five and general	After peak	2.87	1.63	0.57	—
Excluding WD and JAE	Fixed window	1.66	2.28	1.37	[0.87, 2.18]
Excluding WD and JAE	After peak	2.41	2.22	0.92	—
Balanced panel	Fixed window	4.31	5.24	1.22	[0.93, 1.59]
Balanced panel	After peak	7.48	5.09	0.68	—

Table 18: Publication growth and journal placement by country.

Country	Base F	Q growth	F growth	T growth	g	Rank
Finland	21.1	2.09	3.21	1.75	+0.43	1
United Arab Emirates [†]	3.0	8.43	6.23	4.66	-0.30	2
Uruguay [†]	1.9	6.59	4.87	3.11	-0.30	3
Czechia	9.4	5.35	3.14	5.22	-0.53	4
Chile	38.9	4.28	2.30	2.33	-0.62	5
South Africa	25.8	4.48	2.37	3.24	-0.63	6
Israel	122.1	1.49	0.74	1.02	-0.70	7
New Zealand	35.6	1.68	0.77	1.51	-0.77	8
Brazil	39.3	4.99	2.20	2.50	-0.82	9
Colombia	15.9	10.48	4.55	7.31	-0.83	10
South Korea	33.7	4.00	1.72	3.00	-0.85	11
Poland [†]	4.2	5.23	2.18	4.60	-0.88	12
Portugal	20.2	3.73	1.50	2.28	-0.91	13
Argentina	24.2	5.01	1.37	2.18	-1.30	14
Egypt [†]	2.5	13.60	3.49	5.31	-1.36	15
Thailand	5.2	8.12	1.89	3.29	-1.46	16
Mexico	45.4	6.34	1.23	2.37	-1.64	17
India	77.1	10.51	1.78	5.23	-1.78	18
Vietnam [†]	2.2	42.24	7.08	14.66	-1.79	19
Malaysia [†]	3.2	11.76	1.64	6.12	-1.97	20
Russia	31.8	14.58	1.70	4.83	-2.15	21
Bangladesh	9.7	17.00	1.83	6.72	-2.23	22
Greece	14.0	2.78	0.30	1.92	-2.23	23
Turkey	20.0	15.07	1.45	4.30	-2.34	24
Kazakhstan [†]	1.2	28.10	1.80	4.04	-2.75	25
Indonesia	12.6	70.94	1.56	22.95	-3.81	26

Table 19: South African author positions in top-five journals, by publication type.

Author	Year	Publication type
Ingrid Le Roux	2004	AER Papers and Proceedings
Isaac Kalonda-Kanyama	2013	AER Papers and Proceedings
Grant Smith	2015	AER Papers and Proceedings
Justine Burns	2022	Regular AER article
Neil Rankin	2022	Regular AER article

Table 20: Research on South Africa in regular top-five articles.

Comparison	Early articles	Late articles	Ratio	95% interval
Fixed window	5	7	1.05	[0.29, 4.18]
After peak	4	6	0.35	—

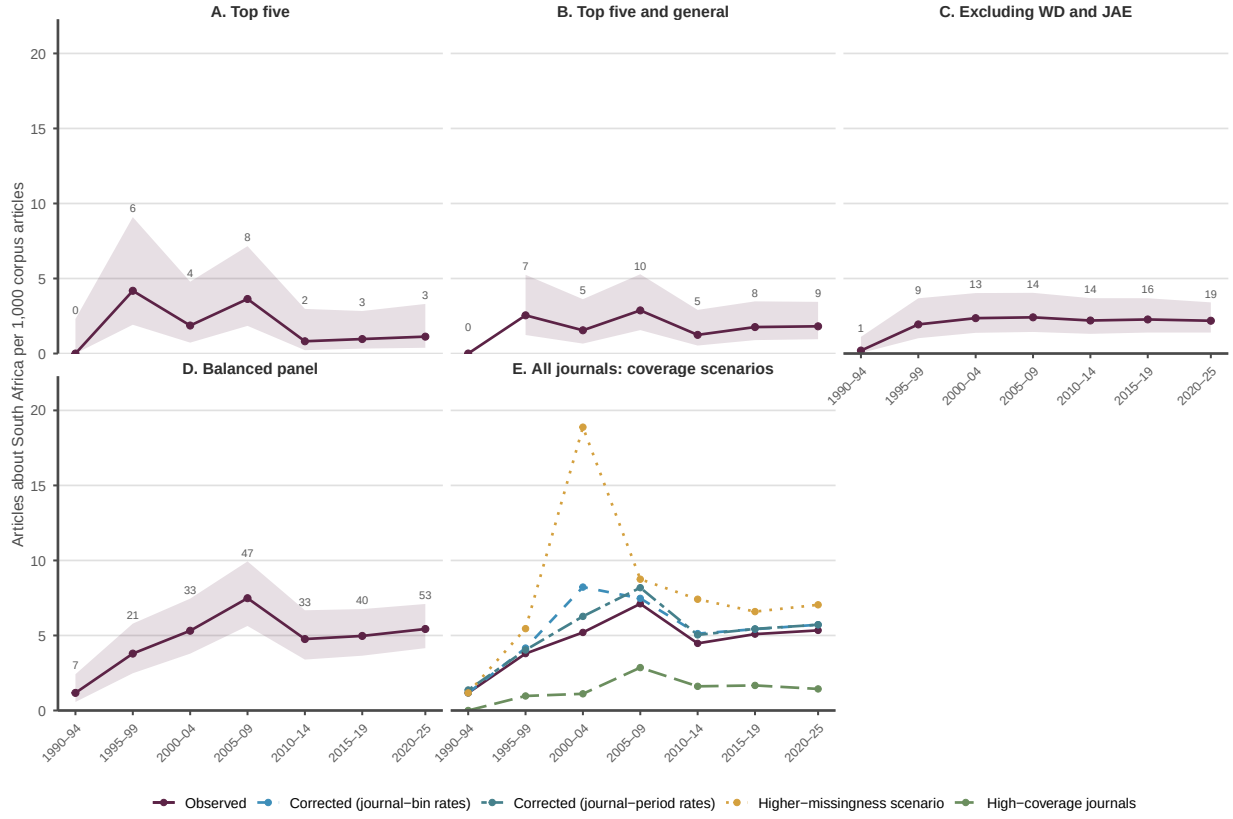


Figure 8: Research on South Africa by journal subset.

Table 21: Descriptive comparisons of firm-topic shares.

Estimator	Estimate (pp)	Placebo rank	Tail fraction	Placebo SD (pp)
Two-way fixed effects, SA x post-2016	-0.09	26 of 26	1.00	3.39
Event study, mean 2020-24 vs 2015	1.72	21 of 26	0.81	4.00
Synthetic control, mean post-2016 gap	-0.02	26 of 26	1.00	3.34
Placebo in time, SA x post-2012 on 2008-15	-1.92	12 of 26	0.46	2.87

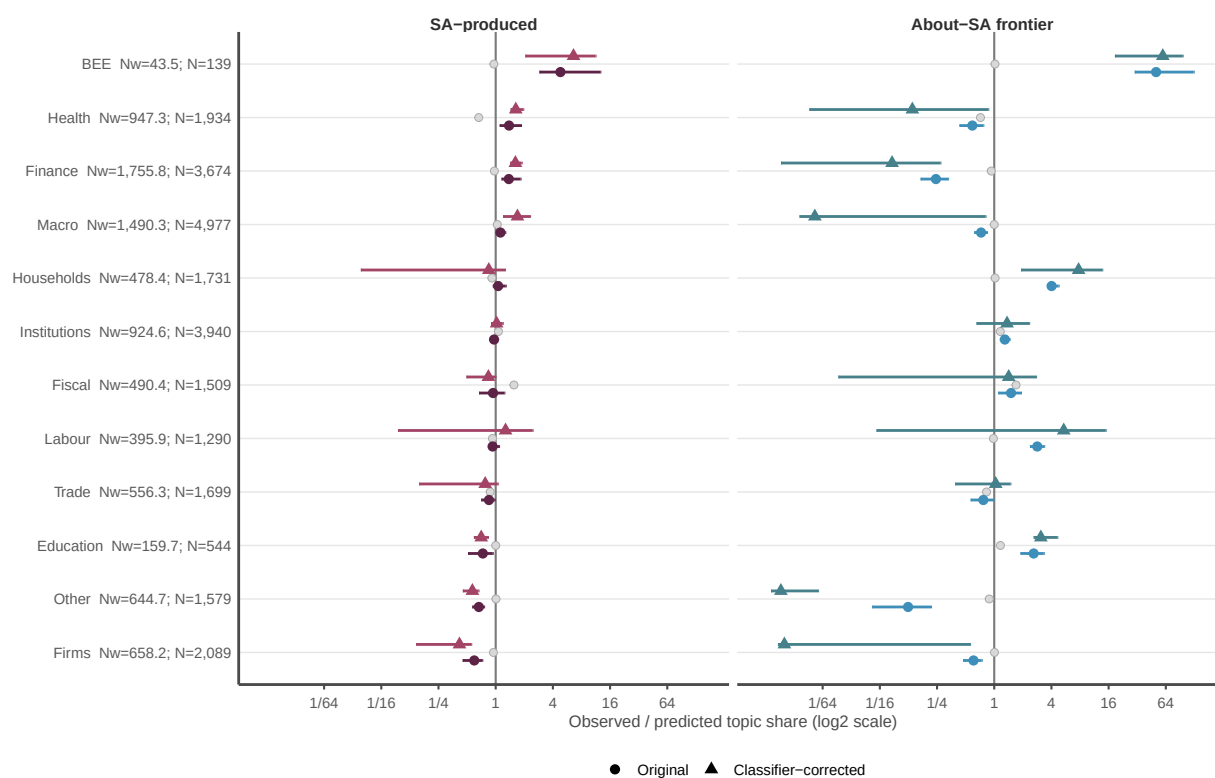


Figure 9: Topic ratios before and after classifier correction.

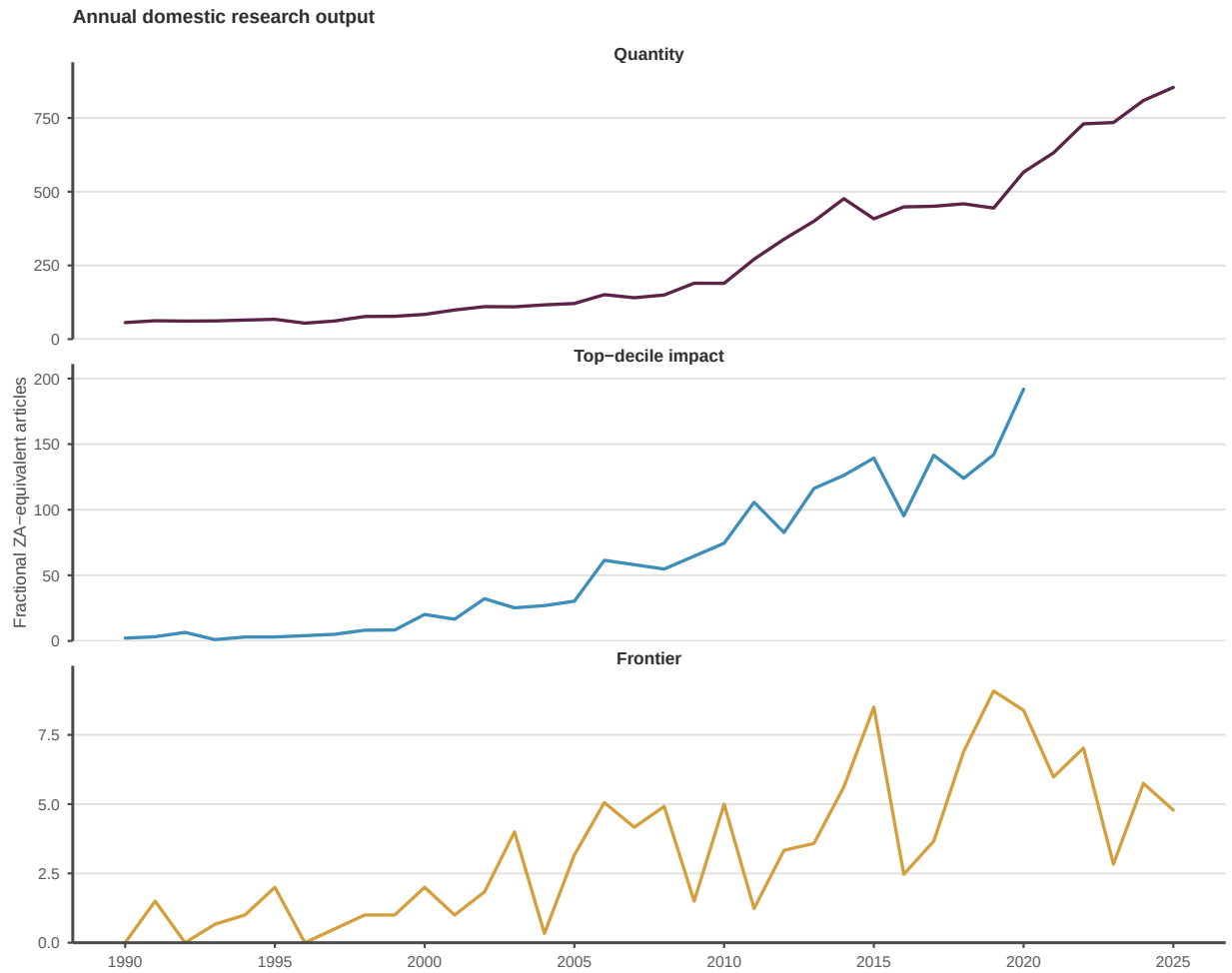


Figure 10: Annual domestic publication, journal placement and citation impact.

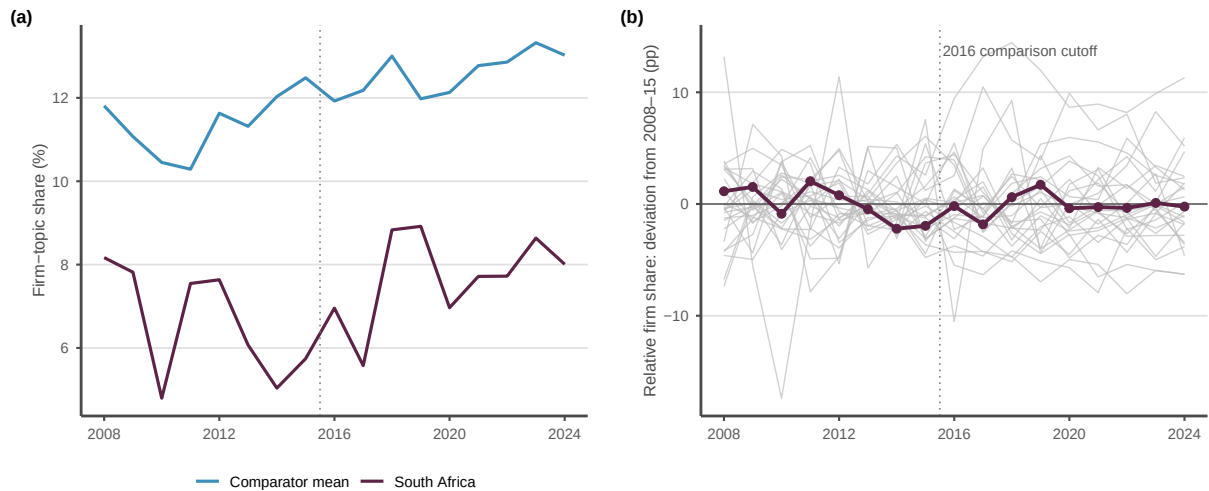


Figure 11: Firm-topic shares in South Africa and comparator countries.